UNIVERSITY OF MESSINA

Department of Mathematical and Computer Sciences, Physical Sciences and Earth Sciences (MIFT)

Bachelor's Degree in Computer Science (L-31) · Data Analysis curriculum



REPORT ON INTERNSHIP ACTIVITIES

RobustCluster-Aβ

Reproducible analysis of amyloid-β40 conformational descriptors

Internship project: Clustering Framework for Intrinsically Disordered Proteins

Student	Ali Erkan Ocaklı (student ID 555012)
Host laboratory	Laboratory of Experimental and Computational Physics of Biosystems, MIFT
Host supervisor	Prof. Valeria Conti Nibali
Co-supervisor	Andrea Ciccolo
Academic tutor	Prof. David Barilla
Hours completed	75

Academic year 2025–2026

Abstract

This report describes a 75-hour curricular internship, completed in full, in the Laboratory of Experimental and Computational Physics of Biosystems at the University of Messina, on the project *Clustering Framework for Intrinsically Disordered Proteins*. After a preparatory study of protein structure, intrinsic disorder and molecular dynamics, with practice in GROMACS and VMD, the final task was to reproduce the clustering workflow of Cadenelli and colleagues for amyloid-β40, vary its parameters, and propose an independent analysis framework.

The supplied table contains 300,003 descriptor observations from three replicas of well-tempered metadynamics. An audit of the supplied script found problems in input handling, residual-cluster treatment, free-energy interpolation and weighted summaries; a modular, tested reimplementation corrects them. Run with the published parameters on all frames, it finds 11 density-peak families, which separate mainly by secondary structure and are shared unevenly between replicas; the partition changes substantially with the initialization seed (ARI 0.44–0.68) and with single parameters. Because the free-energy surface does not state its unit, populations were computed assuming $\mathrm{kJ\,mol^{-1}}$: reweighting removes the helix-rich families and concentrates 77.7% of the weight on three compact or extended families, a pattern that holds even if the file were in $\mathrm{kcal\,mol^{-1}}$.

As the independent framework, Gaussian mixtures were fitted with and without solvent-accessible surface area (SASA), which correlates strongly with radius of gyration ($r = 0.848$): fifty converged fits across two feature sets, five component counts and five seeds. Median agreement between feature sets was 0.864 at three components but 0.457 at six, so whether SASA is redundant depends on the model. Each 100,001-frame series yields only 3.2–28.7 effective samples, so all results are descriptive statements about a biased, correlated sample.

Keywords: curricular internship; amyloid-β40; intrinsically disordered peptides; well-tempered metadynamics; clustering; Gaussian mixtures; reproducibility.

Internship at a glance

THE INTERNSHIP

- **Host.** Laboratory of Experimental and Computational Physics of Biosystems, MIFT, University of Messina.
- **Supervision.** Prof. Valeria Conti Nibali (host supervisor), Andrea Ciccolo (co-supervisor), Prof. David Barilla (academic tutor).
- **Workload and period.** 75 hours completed; registered period 25 May – 30 September 2026; training project approved 25 June 2026.
- **Tools.** GROMACS, VMD, Python (NumPy, SciPy, pandas, scikit-learn, Matplotlib), pytest, Snakemake, Git/GitHub, LaTeX.

KEY FINDINGS

- **The supplied data pass every structural check.** 300,003 frames form exactly three replicas of 100,001 frames at 20 ps; the recovered FES is a complete grid containing every descriptor query.
- **The published clustering is reproduced, but it is sensitive.** With the published parameters, all frames form 11 families; changing only the seed gives ARIs of 0.44–0.68 with that partition.
- **Reweighting reshapes the families (assumed FES unit).** Under the kJ mol^{-1} assumption, three families carry 77.7% of the weight and the helix-rich families almost vanish; the pattern survives the kcal mol^{-1} alternative.
- **Saved frames are far from independent.** Each 100,001-frame series yields only 3.2–28.7 effective samples, and all twelve estimates fail the reliability screen.
- **Correlation does not make SASA redundant.** Dropping SASA preserves the three-component partition (ARI 0.862–0.868; 95.3% of frames keep matched labels), but at six components median agreement falls to 0.457.
- **A single fit can mislead.** At three components seed-to-seed ARI is 0.988–1.000; at two it drops as low as 0.011, which is why every setting was repeated with five seeds.

Abbreviations

Term	Meaning	Term	Meaning
ACF	Autocorrelation function	IDP	Intrinsically disordered protein or peptide
ARI	Adjusted Rand index	MD	Molecular dynamics
CV	Collective variable	PCA	Principal component analysis
DPC / DPA	Density Peak Clustering / Density Peaks Advanced	SASA	Solvent-accessible surface area
ESS	Effective sample size	SEM	Standard error of the mean
FES	Free-energy surface	WT-MetaD	Well-tempered metadynamics
GMM	Gaussian mixture model	MCP	Model Context Protocol

Contents

Abstract i

Internship at a glance ii

1 Introduction 1
- 1.1 Context . . . 1
- 1.2 Internship task . . . 1
- 1.3 Planned and actual work . . . 1
- 1.4 Structure of the report . . . 1

2 Host institution and internship framework 3
- 2.1 Host laboratory and supervision . . . 3
- 2.2 Organization and administrative procedure . . . 3
- 2.3 Programme and timeline . . . 3

3 Scientific and theoretical background 5
- 3.1 Amyloid-β40 and intrinsically disordered proteins . . . 5
- 3.2 Molecular dynamics and the sampling problem . . . 5
- 3.3 Well-tempered metadynamics and free-energy surfaces . . . 5
- 3.4 Statistical reweighting . . . 6
- 3.5 Descriptors and density-based clustering . . . 6

4 Tools, data and methods 7
- 4.1 Tools and technologies . . . 7
- 4.2 Supplied data . . . 7
- 4.3 Reproduction workflow . . . 8
- 4.4 FES interpolation and statistical weights . . . 8
- 4.5 Diagnostics for the independent analysis . . . 9
- 4.6 Reproducibility . . . 9

5 Activities carried out 10
- 5.1 Preparatory study . . . 10
- 5.2 Practical work with GROMACS and VMD . . . 10
- 5.3 Task 1: running and auditing the supplied script . . . 10
- 5.4 Task 2: varying the parameters . . . 11
- 5.5 Task 3: an independent analysis framework . . . 12
- 5.6 Software practice . . . 12

6 Results 14
- 6.1 What the available data support . . . 14
- 6.2 Shared variation in the descriptors . . . 14
- 6.3 Dependence between saved observations . . . 14
- 6.4 Density-peak families in the real data . . . 15
- 6.5 Weighted family populations under an assumed FES unit . . . 16
- 6.6 What changes when SASA is removed . . . 18
- 6.7 Interpretation . . . 19

7 Skills acquired and critical reflection **20**
7.1 Objectives and outcomes . . . 20
7.2 Skills acquired . . . 20
7.3 Related personal projects . . . 20
7.4 Critical reflection . . . 21

8 Conclusions and future work **22**

Signatures **23**

Bibliography **24**

A Full parameter table **26**
Completed descriptor-only GMM pilot . . . 27

B Mathematical derivations and estimator checks **28**

C Additional diagnostics and interpretation **29**

D Code audit summary **33**

E Reproducibility instructions and software environment **35**
Software environment . . . 35

F Personal projects and source-code evidence **36**

1 Introduction

This chapter states the internship task, how the work departed from the original proposal, and how the report is organized.

1.1 Context

Molecular descriptors form a familiar data-analysis problem: clustering groups rows with similar measurements. Interpretation requires care, because consecutive observations may be correlated, simulation bias changes configuration frequencies, and descriptors can hide structural differences. An intrinsically disordered peptide such as amyloid-β40 (Aβ40) is described by many configurations rather than one fixed structure [1], and clustering is a natural way to summarize that diversity.

Cadenelli and colleagues studied monomeric Aβ40 and Aβ42 ensembles with well-tempered metadynamics (WT-MetaD), density-peak clustering, consensus between local clusters and statistical reweighting [2]. Two of the authors supervised this internship, and their clustering script, descriptor table and free-energy surface (FES) for Aβ40 became its working material.

1.2 Internship task

After the preparatory study described in Chapter 2, the supervisors assigned the final task on 18 September 2026 in three parts:

1. **Reproduce.** Run the supplied script and reproduce the published results.
2. **Vary.** Modify its parameters and observe the effects.
3. **Propose.** Independently propose and explore another clustering framework, or other analysis procedures learned during the Bachelor's programme. The supervisors described this as the most important part.

These parts translate into three working questions: can the analysis be reconstructed and tested; what do the descriptors reveal about redundancy and temporal dependence; and does removing the solvent-accessible surface area (SASA), the descriptor most correlated with compactness, preserve the fitted groups?

1.3 Planned and actual work

The published internship proposal described a broader project, in which all-atom simulations would be run with GROMACS and the effect of different solvents analysed. The work actually carried out used the supplied descriptor data, as the 18 September assignment specified; no new simulations or solvent comparison are reported. Within that scope, all three parts of the task were carried out: the published clustering was reproduced on the full data and its parameters were varied (Sections 5.3 and 5.4), and an independent framework was explored (Section 5.5). The weighted family populations were computed under an assumed FES unit, because the supplied file does not document it; they are reported with that assumption and with the effect of the alternative unit.

1.4 Structure of the report

Chapter 2 describes the host laboratory, supervision and programme. Chapter 3 summarizes the scientific background needed to read a biased simulation, and Chapter 4 the tools, data and methods. Chapter 5 gives a detailed account of the activities carried out, block by block, and Chapter 6 reports

the completed analyses. Chapter 7 reflects on the skills acquired against the objectives, and Chapter 8 concludes.

Five limitations apply to the whole report; they are stated here once and referred to later as *Scope 1–5*.

SCOPE AND LIMITATIONS

1. **Sampling.** The descriptors come from WT-MetaD, so observed frequencies are biased and trajectory time is biased time. No equilibrium probabilities or kinetic rates are inferred from them.
2. **Weights.** The supplied FES is a complete, numerically valid grid, but its energy unit and replica-combination recipe are undocumented. Weighted populations are computed under an explicit assumption (kJ mol^{-1}, three replicas combined) and are marked as such (Section 6.5).
3. **Interpretation.** Clusters and Gaussian components are geometric groupings of four descriptors; none is identified as a metastable state or given biological meaning.
4. **Uncertainty.** Frames are strongly autocorrelated; reported minimum–maximum ranges describe the runs performed and are not confidence intervals.
5. **Evidence.** A passing test shows that code is correct on inputs with known answers; it is not an Aβ40 result.

2 Host institution and internship framework

This chapter describes where the internship took place, who supervised it, and how the programme unfolded.

2.1 Host laboratory and supervision

The internship was hosted by the Laboratory of Experimental and Computational Physics of Biosystems of the Department of Mathematical and Computer Sciences, Physical Sciences and Earth Sciences (MIFT) at the University of Messina, on the project *Clustering Framework for Intrinsically Disordered Proteins*. Table 1 gives the administrative frame recorded in the signed training project.

Table 1 Administrative frame of the internship, as recorded in the signed training project and the published internship proposal.

Item	Entry
Student	Ali Erkan Ocaklı, student ID 555012; Computer Science (L-31), Data Analysis curriculum, enrolled 2023/2024, third year in 2025/2026
Host	University of Messina, Department MIFT; Laboratory of Experimental and Computational Physics of Biosystems
Host supervisor	Prof. Valeria Conti Nibali
Co-supervisor	Andrea Ciccolo, PhD student in Physics
Academic tutor	Prof. David Barilla
Workload	75 hours, completed; registered period 25 May – 30 September 2026
Approval	Training project signed by the degree coordinator on 25 June 2026

Prof. Valeria Conti Nibali supervised the project and Andrea Ciccolo, a PhD student in physics, followed the day-to-day work as co-supervisor. Both are co-authors of the Aβ40/Aβ42 study whose clustering script and data form the case study of this report [2], so the final assignment came directly from ongoing research rather than from a textbook exercise. The signed project states the aim as developing computational and analytical skills for the study of intrinsically disordered proteins, with a focus on the amyloid-β peptide.

2.2 Organization and administrative procedure

Study could be done remotely, with regular meetings to discuss the material; the co-supervisor circulated reading lists, lecture material and the analysis files by email, and meetings were held in person or online. Before any activity could count towards the internship, the training project had to be signed by the student, the host supervisor and the academic tutor, approved by the degree coordinator and registered with the university protocol office. At the end, the regulation requires a report on the activities, a signed activity log and the host tutor's assessment form.

2.3 Programme and timeline

The programme moved from biochemistry to simulation and finally to an open analysis task. Table 2 lists the blocks as assigned in the supervisors' emails, and Figure 1 places them in time.

The clustering-project block defines the structure of the remaining chapters. Reproduction and parameter study correspond to Sections 5.3 and 5.4; the independent framework is described in Section 5.5, and its results in Chapter 6.

Table 2 Internship programme as assigned in the supervisors' emails. Dates mark when each block was assigned; the practical tool work is described in Section 5.2.

Block	Content
Proteins and disorder assigned 25 June 2026	Amino-acid structure and classification, ionization, peptide bonds; levels of protein structure, peptide-group conformation, folding and stability; intrinsically disordered proteins. Discussion meeting on 30 June.
Simulation assigned 1 July 2026	Force fields, molecular dynamics, protein dihedral angles and cis/trans conformations; introductory material on setting up and running an MD simulation. Practical work with GROMACS and VMD.
Clustering method resumed 4 September 2026	The supervisors' clustering framework, introduced before the summer break, studied through the Methods and Results sections on conformational clustering of the primary article, with background on density-peak and agglomerative clustering; online walkthrough of the Python implementation on 10 September.
Clustering project assigned 18 September 2026	(1) Run the supplied script and reproduce the published results; (2) vary its parameters and observe the effects; (3) independently propose and explore another clustering or analysis framework from the Bachelor's programme. Progress meetings on 16 and 24 September.
Report	A written report of about 20 pages (minimum 15) presenting the work.

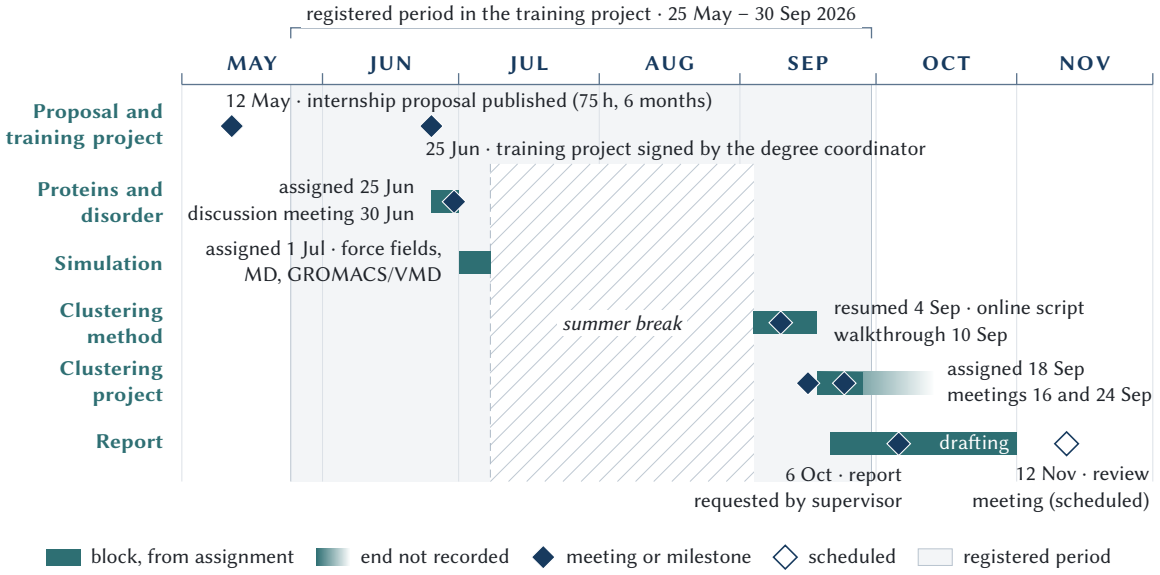


Figure 1 Internship timeline, May–November 2026. The grey band is the period registered in the training project. Teal bars run from each block's assignment to the next assignment; the faded bar marks an end date that was not recorded, and the start of the summer break is approximate. Diamonds are meetings or milestones; the hollow diamond is the report review meeting scheduled for 12 November.

3 Scientific and theoretical background

This chapter summarizes the physics and statistics needed to read a biased simulation correctly.

3.1 Amyloid-β40 and intrinsically disordered proteins

Aβ40 is an intrinsically disordered peptide: no single structure describes it, and its state is an ensemble of conformations together with their statistical weights [1, 2]. Observing a compact and an extended conformation therefore says nothing about how probable either is. Experiments usually report ensemble averages, and simulations can be combined with NMR or SAXS data when the observables are compatible [3].

3.2 Molecular dynamics and the sampling problem

Molecular dynamics integrates the equations of motion of all atoms, with forces from a force field; the primary study used GROMACS, the CHARMM36m force field and PLUMED [2, 4–6]. At equilibrium a configuration with potential energy $U(q)$ contributes in proportion to $\exp[-\beta_T U(q)]$, with $\beta_T = 1/(k_B T)$ per molecule or $1/(RT)$ per mole, so the energy unit must be known before any weight is computed. Saved frames are not independent observations: saving more often adds similar rows, not information, and treating dependent frames as independent makes an estimate look more precise than the sampling supports [7, 8].

3.3 Well-tempered metadynamics and free-energy surfaces

Metadynamics accelerates exploration by adding a history-dependent bias, built from Gaussian hills (Appendix B), along a few collective variables (CVs). In the well-tempered variant the bias grows ever more slowly, and the bias factor $\gamma = (T + \Delta T)/T$ sets its long-time limit [9]:

$$V(s, t \to \infty) = -\left(1 - \frac{1}{\gamma}\right) F(s) + C. \tag{1}$$

For the primary study ($T = 310$ K, $\gamma = 10$) the bias offsets 90% of the free energy, so the sampled CV distribution, $p_b(s) \propto \exp[-F(s)/(\gamma k_B T)]$, is a tenfold-flattened version of the equilibrium one. This is why the method explores efficiently, and why its raw frequencies must be reweighted before they can be read as populations; biased trajectory time does not give unbiased rates.



(a) A conformational ensemble **(b)** Accumulating sampling bias

Figure 2 Conceptual editorial illustrations of two ideas used in the analysis. Flexible ribbons suggest conformational diversity; layered deposits suggest the accumulated bias in Equation (7). The illustrations contain no data.

The free energy of a CV density is $F(s) = -\beta_T^{-1} \log p(s) + C$; integrating out all other coordinates can merge different atomic arrangements at one point of the surface. PLUMED's `sum_hills` writes either the free energy or the negative bias, which differ by $\gamma/(\gamma - 1) \approx 1.11$ for $\gamma = 10$ [10]: a grid

can be numerically perfect and still be the wrong quantity. The offset-corrected bias `metad.rbias` is meaningful only together with the METAD settings that produced it [11].

3.4 Statistical reweighting

Weighting each sampled frame by $\exp(\beta_T V)$ undoes a quasi-static bias. Combined with Equation (1), this gives the final-bias weights used in the primary study:

$$\ell_i \equiv \log w_i = -\frac{1 - 1/\gamma}{RT} F(\alpha_i, \beta_i). \tag{2}$$

At 310 K, $RT = 2.577\ \mathrm{kJ\,mol^{-1}}$, so with $\gamma = 10$ each $\mathrm{kJ\,mol^{-1}}$ of F lowers $\log w$ by 0.349; if the same numbers were $\mathrm{kcal\,mol^{-1}}$, every log weight would be 4.184 times larger. An additive constant in F cancels on normalization, but the unit does not, which is why it matters (Scope 2). Time-dependent estimators such as that of Tiwary and Parrinello instead use $\ell_i = \mathrm{rbias}_i/(RT)$ [12–14].

3.5 Descriptors and density-based clustering

The four clustering descriptors summarize secondary structure (n_α, n_β: continuous CVs that approximate helical and β residue counts, not DSSP assignments [15]), compactness (R_g) and solvent exposure (SASA). They differ in unit and spread, so their scaling is part of the method: in a raw-scale distance R_g counts about twenty times less than the others.

In density-peak clustering [16], a cluster centre combines a high local density ρ_i with a large distance δ_i to any denser point. The published workflow searches for such centres in random subsets and keeps clusters that are large enough and have a sufficient mean silhouette, $(b - a)/\max(a, b)$, where a is a frame's mean distance to its own cluster and b to the nearest other one [2, 17]. Consensus then merges the local clusters into families, every frame is assigned to a family, and statistical weighting decides how much each family contributes.

4 Tools, data and methods

This chapter describes the tools, the supplied data and the methods used in the activities of Chapter 5.

4.1 Tools and technologies

Table 3 lists the software used during the internship and its role. The simulation tools belong to the practical part of the programme; the analysis, testing and reporting tools produced every result in this report.

Table 3 Tools and technologies used during the internship.

Tool	Role in the internship	Where
GROMACS	Molecular-dynamics engine studied in the simulation block; also the engine of the primary study [4]	Section 5.2
VMD	Visualization and structural measurement of molecular systems	Section 5.2
Python 3.12 with NumPy, SciPy, pandas, scikit-learn, Matplotlib	Data validation, clustering, diagnostics, Gaussian mixtures and figures [18–20]	Chapters 5 and 6
pytest and Ruff	Unit and end-to-end tests (38 passing) and static checks	Section 5.3
Snakemake	Workflow definition reusing the package functions [21]	Appendix E
Git, GitHub and GitHub Actions	Version control, public release and continuous integration	Section 5.6
LuaLaTeX, Biber	Typesetting of this report	—

4.2 Supplied data

The analysis starts from a supplied text table, not from atomic coordinates: 300 003 rows with time, an offset-corrected bias field and eight structural measurements. The script, the table and the FES are identified by SHA256 hashes. The published method is the scientific reference [2]; the supplied script provides implementation details, some of which differ from that description and are therefore checked.

A decrease in the time column marks a new replica. Validation requires exactly three segments with strictly increasing times, finite entries and complete coverage of the 1–3 µs interval (times read as picoseconds, as in the script and the article), giving 100 001 frames per replica at 20 ps; malformed rows are rejected, not skipped. Every row keeps its file line, replica and frame index, so that partitions from different runs can be compared frame by frame. The first microsecond is not in the table, so its equilibration cannot be reassessed.

The clustering features are n_α, n_β, R_g (nm) and SASA (nm^2); the other columns are kept for characterization. They are scaled by pooled min–max normalization, $z_f = (x_f - a_f)/(b_f - a_f)$ with the minimum and maximum over the whole interval, so that all replicas and discovery groups share one coordinate system.

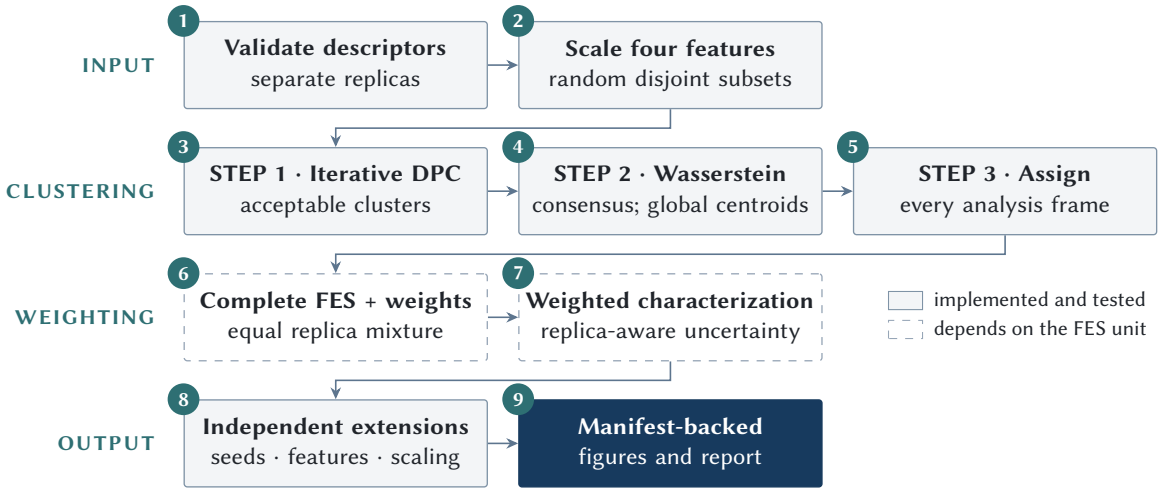


Figure 3 Implemented reproduction workflow. Accepted local clusters define global families, and every frame then receives a label. Dashed stages depend on the FES and were run under the assumed energy unit of Section 6.5.

4.3 Reproduction workflow

Figure 3 summarizes the implemented reproduction. Frames are divided into random, non-overlapping groups of about 30 000, drawn from a recorded master seed [22]. Within a group, the local density ρ_i counts neighbours inside the cutoff d_c, and δ_i is the distance to the nearest point of higher density, ties being broken by frame order. Candidate centres are ranked by $\rho_i\delta_i$, and clusters are accepted only if they reach a minimum size and mean silhouette; the iterative protocol and its parameters are listed in Appendix A. Distances are stored as a blockwise float32 matrix, which limits memory but still grows as N^2.

Consensus. Accepted local clusters are compared with the supplied-code distance

$$D(A,B) = \left[\sum_{f=1}^{4} W_1(A_f, B_f)^2\right]^{1/2}, \tag{3}$$

where each W_1 compares the distributions of one descriptor [23]; it is not a joint four-dimensional transport distance. The baseline keeps the script's raw scales, in which R_g (pooled spread 0.26 nm) counts about twenty times less than n_α, n_β and SASA (5.3, 6.0 and 4.8 nm^2); a normalized variant tests whether this matters. Average-linkage clustering is cut with the script's spline elbow heuristic.

Final assignment. Every frame, including those left unassigned during discovery, is assigned to the nearest family centroid in the normalized space, without iterative refinement.

4.4 FES interpolation and statistical weights

The FES reader checks the grid structure, and descriptor queries $\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$ are interpolated bilinearly with strict bounds [19, 24]; a query outside the grid is an error, not a value to clip. The final-bias log weights of Equation (2) are normalized within each replica before pooling,

$$\widetilde{w}_{ri} = \exp\left[\ell_{ri} - \operatorname{logsumexp}_j(\ell_{rj})\right], \qquad W_{ri} = \widetilde{w}_{ri}/3, \tag{4}$$

so that each replica carries one third of the total weight.

4.5 Diagnostics for the independent analysis

Temporal correlation. The sampling diagnostic asks how much information repeated observations add. For each replica and descriptor, the mean is removed before FFT-based autocorrelation with common $1/N$ covariance normalization. Following Geyer's initial monotone sequence estimator [25], autocorrelations are summed in adjacent pairs, here starting from lag 1. The sum stops at the first non-positive pair, and a running minimum makes the retained pairs non-increasing. The retained contributions define

$$g = \max\left(1, 1 + 2 \sum_{k \in \mathscr{K}} \hat{\rho}(k)\right), \qquad N_{\mathrm{eff,corr}} = N/g, \qquad \tau_{\mathrm{int}} = g\Delta t/2. \tag{5}$$

Here $\mathscr{K}$ identifies the lag contributions retained after this processing; τ_{int} follows the convention $\Delta t(\frac{1}{2} + \sum \rho)$, which differs from the $(g-1)/2$ convention of Chodera [7] by a negligible $\Delta t/2$. Reliability checks require a stopping lag, $N \geq 50g$ (equivalently $N_{\mathrm{eff}} \geq 50$), and a half-window mean difference no greater than 0.25 full-series standard deviations. Geyer's justification assumes a stationary, reversible chain, which a WT-MetaD trajectory is not, so the result serves as a screening diagnostic rather than a formal stationarity test [8].

Partition agreement. ARI compares partitions independently of their label names [26]. It equals 1 for identical partitions, has expected value 0 for random labellings with the same cluster sizes, and can be negative. Matching families by maximum overlap then supports Jaccard and centroid comparisons while retaining unmatched families. Seed variation measures algorithmic sensitivity, not uncertainty between independent replicas. The same comparison is used for the density-peak parameter study (Section 5.4) and for the Gaussian-mixture experiment (Section 5.5).

Feature redundancy. Pearson correlation and PCA of standardized descriptors describe the observed feature geometry, not equilibrium relevance.

4.6 Reproducibility

Every command first writes a manifest with the configuration, input hashes, seeds and software versions, and records runtime and peak memory; the command-line interface and the Snakemake workflow call the same functions [21, 27]. Every figure in this report can therefore be traced to one recorded calculation.

5 Activities carried out

This chapter gives a detailed account of the activities carried out, following the programme blocks of Table 2.

5.1 Preparatory study

The internship required more than applying a familiar statistical method to a new table. The assigned programme began with the concepts needed to interpret the data: amino acids, peptide bonds, protein structure and intrinsic disorder. Force fields, molecular dynamics and dihedral angles connected those concepts to simulation. The final task brought them together through examination of a clustering analysis and an independent investigation.

Two ideas from this block shaped all later work. Saved configurations need not be independent observations, and their frequency in a biased simulation differs from their equilibrium probability. Both connect the molecular setting to statistics and explain why the origin of the data matters as much as the clustering method.

5.2 Practical work with GROMACS and VMD

I used GROMACS, VMD and Python during the internship. Learning to use GROMACS was one of the more challenging parts of the internship for me. I found the process demanding at first, but I also enjoyed becoming more familiar with the software. The simulation block introduced the practical steps of setting up and running a molecular-dynamics simulation, and VMD was used to inspect and measure molecular structures. These exercises were training activities: the Aβ40 data analysed in this report were produced by the original study, not by simulations run during the internship. Working with both programs also led to two personal software projects that wrap them in a simpler interface (Section 7.3). Figure 16 in Appendix F places the tools in relation to each other and to these projects.

5.3 Task 1: running and auditing the supplied script

Running the supplied script

The audit separated three questions: what the paper describes, what the supplied code attempts, and what execution records show. The supplied script combines input reading, replica separation, clustering, weighting and plotting in one file, and a bare Italian sentence at line 437 stops it with a syntax error before anything runs. No original execution record was available, so the script is evidence of intent rather than of a completed run; it is preserved unchanged.

Identified methodological issues

The corrections in Table 4 concern which observations enter the analysis and how clusters are interpreted.

Random grouping was also corrected: the script draws its groups from a fixed list of 17 seeds, which runs out if more groups are requested, whereas the reimplementation derives independent streams from one recorded seed.

Table 4 Methodological corrections in the modular implementation. The original script remains unchanged.

Supplied-script behaviour	Correction and consequence
Takes the first three time-reset segments and skips some malformed rows.	Checks every row and requires exactly three segments, making incomplete or invalid input an explicit failure.
Sends rejected frames into consensus under a generic label.	Uses accepted memberships to discover global centres. Rejected frames receive final labels without being treated as a discovered family.
Permits missing FES cells; nearest-grid lookup can return a boundary value for an out-of-range query.	Requires a complete grid and bounded bilinear interpolation; queries outside the grid fail.
Weights relevance but uses unweighted descriptor means and standard deviations.	Weights cluster summaries consistently and normalizes each replica before pooling. Distribution width, replica SEM and bootstrap intervals remain distinct.

Verification and numerical tests

The 38 tests check each numerical operation on inputs with known answers, such as an analytic bilinear surface and weight normalization with unequal replica lengths, and run synthetic end-to-end fixtures; two worked examples are given in Appendix D.

Remaining ambiguities

The paper does not fully specify the scaling used in consensus, how the replica-averaged FES was built, or the conventions behind every reported uncertainty, and an intermediate script version mentioned in the materials could not be located. Where the code was clearer, its choice was adopted and recorded as source-informed.

Running the corrected workflow on the real data

The three clustering steps (local density-peak discovery, Wasserstein consensus and final assignment) do not use statistical weights: only the final characterization of families needs the FES. A separate `cluster` command therefore runs these steps without reading the FES, and reports family sizes as raw frame fractions of the biased sample rather than populations. With the published parameters and group size, all 300 003 frames entered discovery in 10 random groups of about 30 000; 96 local clusters passed the size and silhouette criteria, and consensus merged them into 11 families (Section 6.4). The run took 14 minutes with a peak memory of 3.8 GiB, measured before the published full profile was enabled. Rerunning with the same seed reproduces the partition exactly. This completes the reproduction of the clustering itself. The weighted populations were then computed under an explicitly stated assumption about the FES unit (Section 6.5).

5.4 Task 2: varying the parameters

With the workflow running on the real data, the parameter study could be carried out. Every variant was compared with the reference partition (seed 42, published parameters) on the same frame identities using the adjusted Rand index. Two kinds of variation were tested, each at the published full profile and at a reduced medium profile (three groups of 10 000 frames):

- *Initialization*: the random assignment of frames to discovery groups, with four further seeds derived from the master seed;
- *One parameter at a time*: the cutoff d_c scaled by 0.9 and 1.1, the silhouette threshold set to 0.25 and 0.35, and the minimum cluster size set to 200 and 400 frames.

A third comparison replaced the raw descriptor scales in the Wasserstein consensus with normalized ones, testing whether the under-weighted R_g changes the families. The results are reported in Section 6.4. Comparisons that still need further inputs were prepared but not run:

- *Time-dependent reweighting*: keep labels fixed and contrast final-bias weights with `metad.rbias` weights [12–14], once the METAD settings are known;
- *Alternative density clustering*: DPA [28]; ShapeGMM would additionally need structural coordinates [29];
- *External validation*: comparison with an experimentally constrained Aβ40 ensemble [3].

5.5 Task 3: an independent analysis framework

The supervisors asked for an alternative framework drawn from the Bachelor's programme. The data-mining course of the curriculum had covered correlation analysis, feature scaling and outlier detection; the independent framework builds on these and extends them with principal component analysis and Gaussian mixture models. The descriptor diagnostics suggested a concrete question for this framework: radius of gyration and SASA are strongly correlated (Chapter 6), so is SASA redundant for clustering? The experiment tests whether removing SASA changes assignments, comparing four descriptors with n_α, n_β and R_g using Gaussian mixtures:

$$p(\mathbf{x}) = \sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \Sigma_j), \qquad \sum_j \pi_j = 1. \tag{6}$$

In Equation (6), components approximate the sampled descriptor distribution; their mixture coefficients are not equilibrium weights. The scikit-learn implementation uses full covariance matrices [20].

The protocol was fixed before execution (Appendix A): every fit uses the same 3000 frames, 1000 per replica; the scaling is estimated on those rows only and shared between the two feature sets; each of the five component counts $K = 2, \ldots, 6$ is fitted with five seeds in both feature spaces, giving 50 fits; and all 300 003 frames are then labelled and compared with ARIs, five paired across feature sets and ten between seeds within each feature set.

The comparison is descriptive: keeping the fitting sample fixed isolates the feature and seed effects, and observed ranges are not confidence intervals.

5.6 Software practice

Small programming decisions also became explicit parts of the analysis. Feature scaling is fitted on a specified sample and reused for later observations. A test checks that a large evaluation value cannot alter that transformation. Original frame identities are retained so that label arrays refer to the same observations. These details determine whether a difference comes from the method under investigation or an unintended change in preprocessing.

Fit once, reuse the same scale RobustCluster-Aβ / Python

```
def fit_scaler(fit_values, values):
    fit_values, values = finite(fit_values), finite(values)
    mean, scale = fit_values.mean(axis=0), fit_values.std(axis=0)
    if np.any(scale <= 0):
        raise InvalidInput("Constant fit feature")
    return (values - mean) / scale, {"mean": mean.tolist(), "scale": scale.tolist()}
```

Figure 15 in Appendix D shows how the package fits together: inputs are validated once at the boundary, analysis functions receive explicit arrays and settings, and each command writes a manifest before producing tables or figures.

6 Results

This chapter reports the completed analyses: data and FES checks, descriptor relationships, temporal dependence, the density-peak families and the feature experiment.

6.1 What the available data support

The calculations reported here use the supplied Aβ40 descriptors. Tests with generated data are discussed with the software validation and are not included as molecular observations. Table 5 distinguishes the completed analyses from the unresolved reproduction steps.

Table 5 Scope of the recorded calculations.

Analysis	Status and scope
Descriptor diagnostics	Complete; unweighted observations from three replicas
FES grid and query checks	Complete; energy unit and construction undocumented
GMM feature comparison	Complete; 50 fits on a fixed fitting sample
Paper DPC clustering	Complete; unweighted, published parameters and full profile
DPC parameter study	Complete; seeds and one-parameter changes
Weighted family populations	Complete under an assumed kJ mol^{-1} unit (Section 6.5)
Structural representatives	BLOCKED: needs coordinates and a frame map

Input validation found three segments of 100 001 rows. Times increase at the source-labelled spacing of 20 ps, and the data satisfy the column, finite-value and interval checks. The descriptor calculation, run `20260921T011959-583b4421`, took 2.28 s with a sampled peak worker-process memory of 418.2 MiB; these figures describe the local diagnostic run, not the cost of a full DPC analysis.

The FES check resolved the earlier incomplete-download problem. The recovered file contains 2,801 by 1,301 grid points, giving 3,644,101 rows with five finite columns. Every expected coordinate pair is present and none is duplicated. The nonperiodic axes cover 0–28 and 0–13 at spacing 0.01, and all 300,003 converted queries fall inside the grid and yield finite bilinear interpolations. Figure 4 shows why this matters: the sampled frames occupy a triangular region of low free energy near small α and moderate β, while 61% of the grid sits on a flat maximum that lies largely outside the sampled region. The file is numerically usable; only its unit stands between it and equilibrium weights (Scope 2).

6.2 Shared variation in the descriptors

The pooled Rg–SASA correlation is 0.848, against -0.456 between the two secondary-structure descriptors, and the relationship holds in every replica (per-replica r between 0.805 and 0.863). It is not a straight line: compact frames ($R_g < 1.3$ nm) form a dense core in which SASA rises steeply with R_g, while extended frames spread along a flatter band (Figure 10 in Appendix C). After standardization the first two principal components explain 51.8% and 36.3% of the variance, together 88.1%. This strong association is a reason to test whether SASA is needed for clustering (Section 6.6), not a reason to drop it in advance.

6.3 Dependence between saved observations

Each replica stays near a compact radius of gyration of about 1.0–1.2 nm for long stretches and leaves it only in a few extended excursions lasting tens to hundreds of nanoseconds of biased time (Figure 9 in Appendix C), so neighbouring frames, 20 ps apart, are nearly copies of each other. For

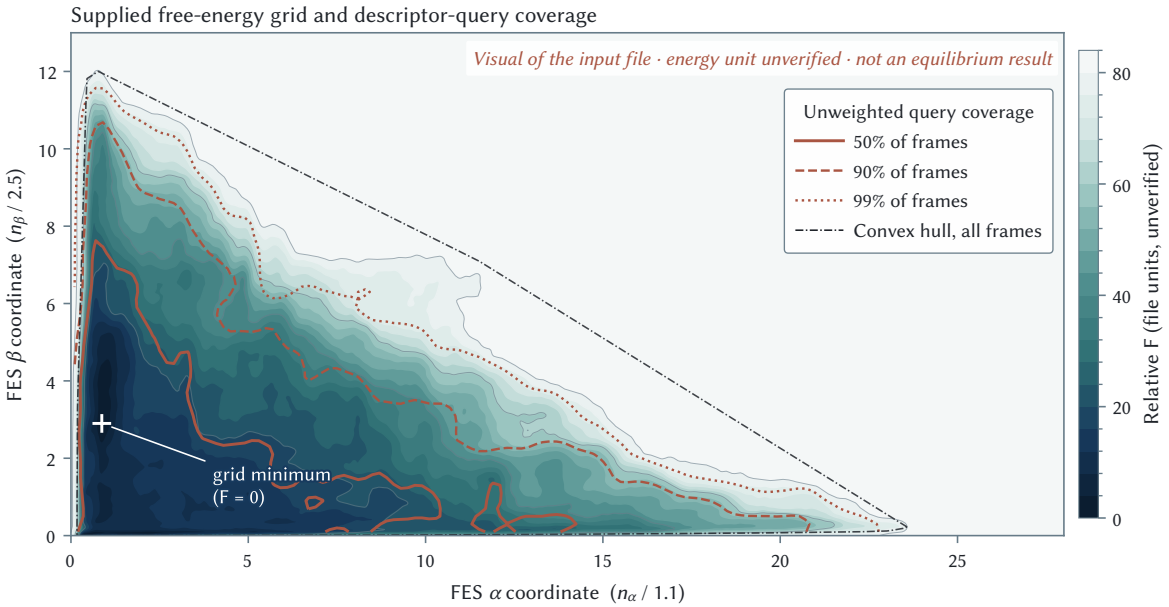


Figure 4 The supplied free-energy grid shifted so its minimum is zero and plotted at one-quarter resolution. Values stay in undeclared file units. Terracotta contours enclose 50%, 90% and 99% of the 300,003 unweighted descriptor queries ($\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$); the dash-dot line is their convex hull, and no query falls outside the grid.

the 12 replica–descriptor series, statistical inefficiencies range from 3,490 to 31,742 frames, giving correlation ESS values of only 3.2–28.7 per 100 001 frames (Table 13). Every series falls below the required $N_{\text{eff}} \geq 50$, and 11 of 12 shift by more than 0.25 standard deviations between their halves; n_α in replica 3 has the longest integrated time, 317 ns of biased time. Independent-frame error bars would therefore be unjustified. These diagnostics describe the supplied 1–3 µs data, not the convergence of the published runs, whose first microsecond and bias history are not available.

6.4 Density-peak families in the real data

The reproduced workflow divides the 300 003 frames into 11 families (Figure 5 and Table 6). The largest holds 15.9% of the frames and the smallest 1.6%. The families separate mainly by secondary structure rather than by compactness. Nine of the eleven have mean radii of gyration between 1.04 and 1.40 nm and overlap in Figure 5A, whereas in the (n_α, n_β) plane of Figure 5B their centres spread from helix-rich ($\bar{n}_\alpha$ up to 16.6) to β-rich ($\bar{n}_\beta$ up to 19.3) regions. Only two families are extended, with mean R_g of 1.58 and 1.81 nm; together they hold 11.5% of the frames.

The replicas do not share the families evenly. Family 3, the most helix-rich, holds 18.8% and 19.5% of the frames of replicas 1 and 3 but only 0.3% of replica 2, while the β-rich family 5 holds 22.9% of replica 2 and 2.9% of replica 3; the largest difference between replicas for a single family is 19.9 percentage points. This agrees with the temporal diagnostics of Section 6.3: each replica spends long stretches in a few regions, so the three replicas sample the landscape in different proportions.

The partition is sensitive to how it is computed (Figure 6). Changing only the initialization seed gives ARIs of 0.44–0.68 with the reference and between 12 and 14 families. Changing one parameter at a time gives ARIs of 0.57–0.74 and between 11 and 14 families. The amount of data entering discovery matters as well: the medium profile, which uses 30 000 frames for discovery instead of all of them, finds only 3 families. Replacing the raw descriptor scales in the consensus with normalized ones left the medium-profile partition unchanged (ARI 1.00), so the low raw-scale weight of R_g does not affect the result there.

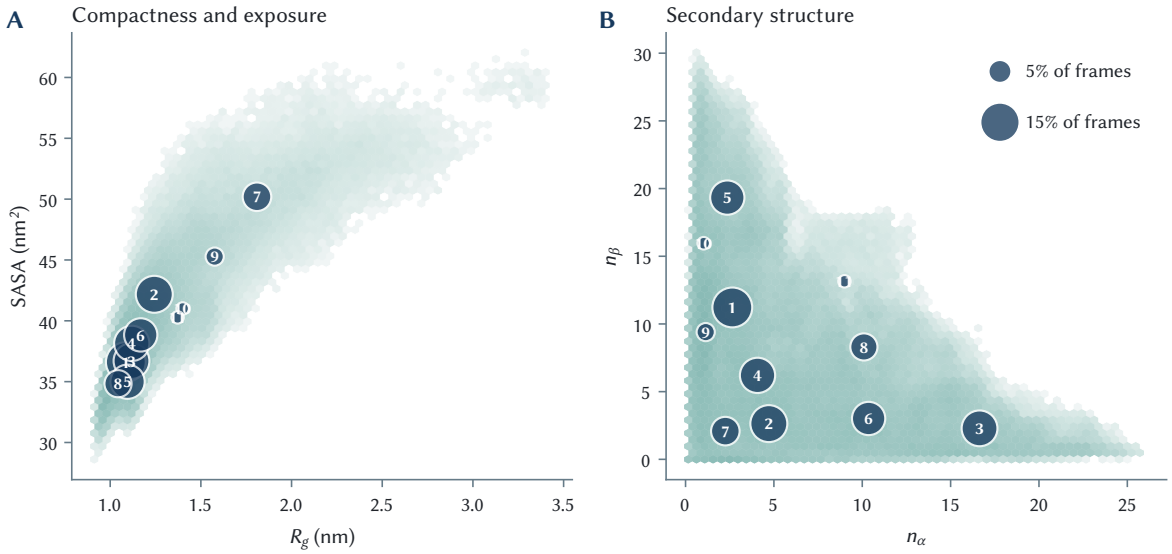


Figure 5 Density-peak families of the reproduced workflow (published parameters, full profile, seed 42) on the unweighted descriptor landscape. Bubbles mark family means, numbered as in Table 6; their area is proportional to the raw frame fraction. The background shows the density of all frames.

Table 6 Density-peak families: raw frame fractions, overall and by replica, and unweighted descriptor means.

Family	Frames	Fraction by replica			$\bar{n}_\alpha$	$\bar{n}_\beta$	$\bar{R}_g$ (nm) / SASA (nm²)
	(all)	1	2	3			
1	15.9%	10.4%	26.1%	11.3%	2.6	11.2	1.09 / 36.6
2	13.7%	14.0%	11.2%	16.0%	4.7	2.6	1.24 / 42.2
3	12.9%	18.8%	0.3%	19.5%	16.6	2.3	1.12 / 36.7
4	12.5%	8.7%	14.6%	14.2%	4.1	6.2	1.12 / 38.1
5	11.8%	9.6%	22.9%	2.9%	2.4	19.3	1.10 / 35.0
6	11.1%	19.1%	4.8%	9.5%	10.4	3.0	1.17 / 38.8
7	8.2%	4.2%	9.3%	11.0%	2.3	2.1	1.81 / 50.2
8	7.4%	11.6%	2.1%	8.6%	10.1	8.3	1.04 / 34.8
9	3.3%	1.0%	4.5%	4.5%	1.1	9.4	1.58 / 45.3
10	1.6%	1.0%	1.3%	2.4%	1.0	16.0	1.40 / 41.0
11	1.6%	1.7%	3.0%	0.0%	9.0	13.2	1.37 / 40.3

Fractions count frames of the biased sample; they are not equilibrium populations (Scope 1–2).

6.5 Weighted family populations under an assumed FES unit

> **ASSUMPTION**
>
> The supplied FES does not state its energy unit or how the three replicas were combined. The populations in this section assume that `FES_AB_40.dat` is the `sum_hills` free energy (`file.free`) of the three replicas combined, in kJ mol^{-1}. This is the PLUMED default, the unit used by the supplied script and the unit of the primary article, but it has not been confirmed by the authors. If the assumption is wrong, these populations are wrong; the kcal mol^{-1} column of Table 7 shows how much they would change.

Under this assumption, the final-bias weights of Equations (2) and (4) were applied to the same 11 families; rerunning the clustering inside the weighted workflow reproduced the unweighted labels

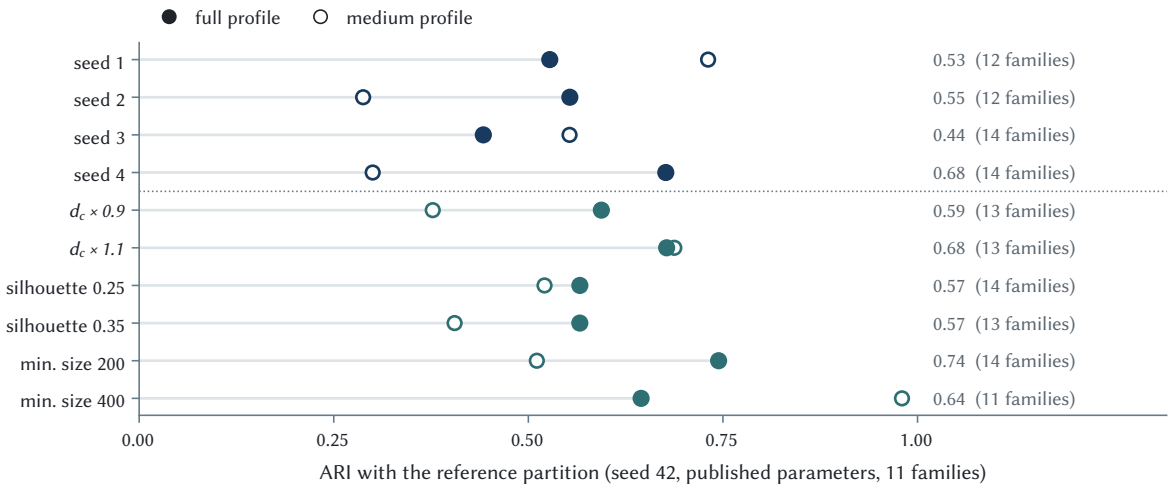


Figure 6 Sensitivity of the density-peak partition. Each row changes either the initialization seed or one parameter of the published configuration; dots give the ARI with the reference partition, filled for the full profile and open for the medium profile, and the labels give the full-profile ARI and family count.

exactly. Reweighting changes the picture substantially (Figure 7 and Table 7). Family 1, compact with little helix and moderate β content, carries 47.4% of the equilibrium weight, and the three largest families together 77.7%. The helix-rich families 3, 6, 8 and 11 and the strongly β-rich family 5, which together hold 45% of the frames, keep less than 1% of the weight: the bias drove the simulations into these regions, but they are rare at equilibrium. The two extended families gain weight, from 11.5% of the frames to 23.8%. The largest change for a single family is 31.5 percentage points.

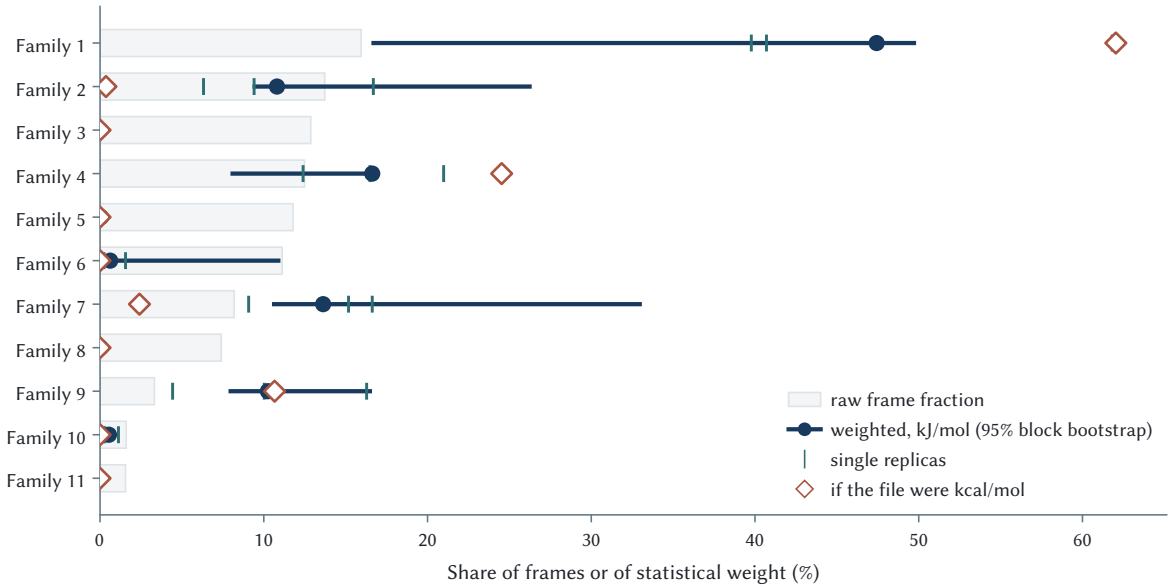


Figure 7 Family populations before and after reweighting, assuming the FES is in kJ mol^{-1}. Grey bars are raw frame fractions; navy dots are weighted populations with rough 95% block-bootstrap intervals; teal ticks are the three replicas; open diamonds show the populations if the same file were in kcal mol^{-1}.

The weights are strongly concentrated. The Kish effective sample size is 3,277–9,551 per replica of 100 001 frames, and the replicas disagree: the population of family 1 ranges from 39.8% to 61.8% across replicas. If the file were in kcal mol^{-1}, every log weight would be 4.184 times larger, and family populations would shift by up to 14.6 percentage points; family 1 would still dominate, with 62.0%, and the families that vanish under the kJ mol^{-1} assumption would vanish as well. The qualitative

Table 7 Raw frame fractions and weighted populations of the density-peak families (%), under the assumed kJ mol^{-1} unit and, for comparison, if the file were in kcal mol^{-1}.

Family	Raw	Weighted	95% bootstrap	Replica range	kcal/mol case
1	15.9	47.4	16.6–49.8	39.8–61.8	62.0
2	13.7	10.8	9.3–26.4	6.3–16.7	0.4
3	12.9	0.0	0.0–0.3	0.0–0.0	0.0
4	12.5	16.6	8.0–17.0	12.4–21.0	24.5
5	11.8	0.1	0.0–0.1	0.0–0.1	0.0
6	11.1	0.6	0.4–11.0	0.1–1.6	0.0
7	8.2	13.6	10.5–33.1	9.1–16.6	2.4
8	7.4	0.0	0.0–0.4	0.0–0.1	0.0
9	3.3	10.2	7.8–16.6	4.4–16.3	10.7
10	1.6	0.5	0.1–0.9	0.2–1.1	0.0
11	1.6	0.0	0.0–0.0	0.0–0.0	0.0

Bootstrap blocks are twice the largest statistical inefficiency of each replica (21,687–63,485 frames), so each replica contributes only two to five blocks and the intervals are rough. They hold the partition and the FES fixed.

result, that reweighting removes the helix-rich and strongly β-rich families and concentrates the ensemble on a few compact and extended families, therefore does not depend on which of the two units is correct; the individual percentages do.

6.6 What changes when SASA is removed

All 50 Gaussian-mixture fits converged; the pilot took 7.01 s with a peak of 660.2 MiB. Paired fits with and without SASA share observations and seed, so their ARI measures the feature effect, while fits within one feature space differ only in initialization (Figure 8; values in Table 12, Appendix C).

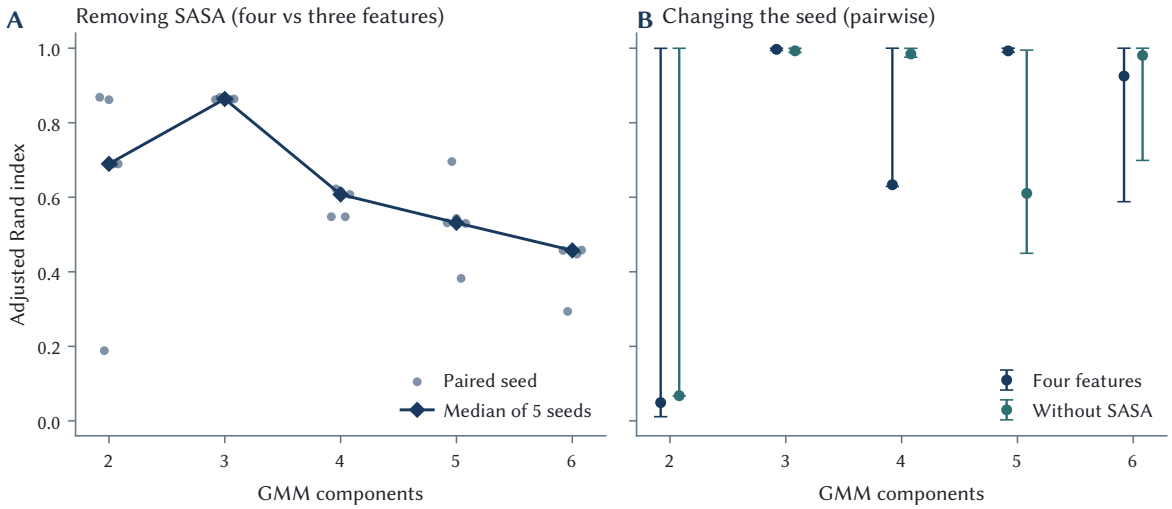


Figure 8 The independent GMM comparison. Left: ARI between four-feature and SASA-removed assignments for five paired seeds; diamonds denote medians. Right: within-feature-space ARI for ten seed pairs. Bars span observed minima and maxima. Every model uses the same fitting subset, and evaluation includes fitting rows.

At three components the cross-feature ARIs are 0.862–0.868, and a median 95.3% of frames keep corresponding labels; within-space agreement is 0.994–1.000 with four descriptors and 0.988–1.000 without SASA. At two components, by contrast, within-space ARIs fall to 0.011 and 0.067, so a single

fit would have been misleading (all seed pairs in Figure 14). Seed stability alone does not single out three components, since four-feature fits at five components (0.989–1.000) and three-feature fits at four (0.976–1.000) are equally reproducible, and within-space BIC keeps decreasing up to six. What distinguishes three components is that the partition is stable in both feature spaces and agrees across them; at the other counts the median cross-feature ARI falls to 0.457–0.690.

6.7 Interpretation

Correlation alone does not settle a feature choice: it summarizes the whole sample, whereas clustering assigns individual frames using all retained features, and the answer here depends on the model. The two clustering approaches also behave differently. The three-component Gaussian mixture is almost insensitive to its seed, whereas the density-peak partition changes substantially with the seed, the parameters and the amount of data used for discovery. The density-peak families should therefore be read as one reproducible outcome of a stated configuration rather than a unique decomposition of the ensemble, and their weighted populations depend in addition on the assumed FES unit.

7 Skills acquired and critical reflection

This chapter compares the outcomes with the internship objectives and reflects on the skills acquired.

7.1 Objectives and outcomes

Table 8 compares the aims of the training project and the three parts of the final task with what was achieved, pointing to the evidence for each.

Table 8 Internship objectives compared with outcomes.

Objective	Status	Evidence
Develop computational and analytical skills for intrinsically disordered proteins (training project)	Achieved	Preparatory study and practice (Chapter 3, Section 5.2); analyses of Chapter 6
Run the script and reproduce the published results	Achieved, one assumption	Script audited and corrected; clustering reproduced on all frames with the published parameters (Section 5.3); weighted populations under an assumed FES unit (Section 6.5)
Vary the parameters	Achieved	Seed, cutoff, silhouette, cluster-size and scaling variants on the real data (Sections 5.4 and 6.4)
Propose and explore an independent framework	Achieved	50-fit Gaussian-mixture feature experiment with PCA and temporal diagnostics (Section 5.5, Section 6.6)
Report of about 20 pages	Achieved	This document

7.2 Skills acquired

Table 9 summarizes the skills exercised during the internship and where each is documented. It records activities rather than self-assessed levels; the evaluation of the internship belongs to the host tutor.

7.3 Related personal projects

Learning GROMACS and VMD during the internship showed me how many commands, options and file formats a newcomer has to keep track of. To make these programs easier to use, I created two personal software projects, VMD MCP and GROMACS MCP. They use the Model Context Protocol to expose molecular-software operations through structured inputs and outputs. Both were written in early July 2026, during the simulation block: `vmd-mcp` exposes 12 tools for structure loading, measurement and display-free rendering, and `gromacs-mcp` exposes 31 tools covering system preparation, simulation runs and analysis, with a tagged v0.2.0 release and continuous integration. They are distinct from the Aβ40 descriptor analysis, but provide further examples of connecting scientific programs with Python interfaces.

Source excerpts, screenshots and a map of the tools are in Appendix F. No Aβ40 data in this report were produced with these projects.

Table 9 Skills exercised during the internship and the evidence in this report.

Skill area	Activity and evidence
Protein structure and intrinsic disorder	Preparatory study; background of Chapter 3
Molecular simulation software	GROMACS and VMD practice (Section 5.2)
Enhanced sampling and reweighting	Reading WT-MetaD output; final-bias weights and their unit dependence (Section 3.4)
Scientific Python and data validation	Replica parsing, FES grid checks, frame identities (Chapter 4 and Section 5.3)
Time-series diagnostics	Autocorrelation and effective sample size (Section 4.5, Chapter 6)
Clustering and model comparison	Density-peak workflow; Gaussian mixtures, ARI, BIC (Sections 5.5 and 6.6)
Software testing and reproducibility	38 tests, run manifests, continuous integration (Sections 5.3 and 5.6)
Scientific writing	This report in LaTeX, with figures generated from saved tables

7.4 Critical reflection

The most demanding part of the internship was practical rather than statistical. Learning to use GROMACS took time, and it made clear how many decisions lie behind a simulated ensemble before any clustering starts. Working with supplied files raised a second kind of difficulty: the script did not run as delivered, an intermediate version was missing, and the free-energy surface did not state its unit. Each of these was recorded and handled explicitly instead of being patched over. Separating the clustering from the weighting step later made it possible to reproduce the published clustering in full while leaving only the populations open.

What went well was turning a single observation into a controlled experiment. The strong Rg–SASA correlation could have justified dropping a feature; testing that decision with fixed samples and repeated seeds showed that the answer depends on the model. A practical lesson for future projects is to ask early for the provenance of every input file, since one missing unit was enough to block the weighting stage.

8 Conclusions and future work

This chapter summarizes what was achieved and what the next step should be.

The internship turned a supplied clustering script and data set into an inspectable, tested analysis. The audit identified problems in parsing, residual-cluster handling, FES interpolation, replica pooling and weighted summaries, and the modular reimplementation corrects them with 38 passing tests while preserving the original sources.

The corrected workflow reproduces the published density-peak clustering on all 300 003 frames. It finds 11 families that differ mainly in secondary structure and are shared unevenly between replicas, and the parameter study shows that the partition depends strongly on the seed, the parameters and the data used for discovery. Under an assumed $\mathrm{kJ\,mol^{-1}}$ FES unit, reweighting concentrates 77.7% of the equilibrium weight on three families and almost removes the helix-rich ones, a pattern that also holds if the file were in $\mathrm{kcal\,mol^{-1}}$. The independent Gaussian-mixture experiment shows that the strong Rg–SASA correlation does not settle the feature choice: removing SASA preserves the three-component partition but not the others.

The most important open point is to confirm the FES unit and how the replicas were combined with the authors; if either differs, the populations of Section 6.5 can be recomputed at once from the saved labels. A comparison with the Aβ40 families reported in the article should then use several seeds, ideally the twenty-seed study the configuration already provides, and the feature experiment should be repeated on disjoint time blocks to address the sampling sensitivity that the fixed fitting sample excludes [8]. With atomic trajectories and a frame map, representative structures and coordinate-based methods such as ShapeGMM would become possible [29, 30].

Signatures

Student Ali Erkan Ocaklı	Host supervisor Prof. Valeria Conti Nibali	Academic tutor Prof. David Barilla
Place and date	Place and date	Place and date

Bibliography

[1] R. van der Lee, M. Buljan, B. Lang, et al. "Classification of Intrinsically Disordered Regions and Proteins". In: *Chemical Reviews* 114.13 (2014), pp. 6589–6631. DOI: 10.1021/cr400525m.

[2] I. Cadenelli, A. Ciccolo, A. Tagliabue, G. Rossi, V. Conti Nibali, and D. Bochicchio. "The Monomeric Conformational Ensembles of Aβ40 and Aβ42 Encode Their Differential Amyloid Aggregation Propensity". In: *The Journal of Physical Chemistry B* 130.24 (2026), pp. 6035–6046. DOI: 10.1021/acs.jpcb.6c01000.

[3] K. Borthakur, T. R. Sisk, F. P. Panei, M. Bonomi, and P. Robustelli. "Determining accurate conformational ensembles of intrinsically disordered proteins at atomic resolution". In: *Nature Communications* 16.1, 9036 (2025). DOI: 10.1038/s41467-025-64098-3.

[4] M. J. Abraham, T. Murtola, R. Schulz, et al. "GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers". In: *SoftwareX* 1-2 (2015), pp. 19–25. DOI: 10.1016/j.softx.2015.06.001.

[5] J. Huang, S. Rauscher, G. Nawrocki, et al. "CHARMM36m: an improved force field for folded and intrinsically disordered proteins". In: *Nature Methods* 14.1 (2017), pp. 71–73. DOI: 10.1038/nmeth.4067.

[6] G. A. Tribello, M. Bonomi, D. Branduardi, C. Camilloni, and G. Bussi. "PLUMED 2: New feathers for an old bird". In: *Computer Physics Communications* 185.2 (2014), pp. 604–613. DOI: 10.1016/j.cpc.2013.09.018.

[7] J. D. Chodera. "A Simple Method for Automated Equilibration Detection in Molecular Simulations". In: *Journal of Chemical Theory and Computation* 12.4 (2016), pp. 1799–1805. DOI: 10.1021/acs.jctc.5b00784.

[8] A. Grossfield, P. N. Patrone, D. R. Roe, A. J. Schultz, D. Siderius, and D. M. Zuckerman. "Best Practices for Quantification of Uncertainty and Sampling Quality in Molecular Simulations [Article v1.0]". In: *Living Journal of Computational Molecular Science* 1.1 (2019). DOI: 10.33011/livecoms.1.1.5067.

[9] A. Barducci, G. Bussi, and M. Parrinello. "Well-Tempered Metadynamics: A Smoothly Converging and Tunable Free-Energy Method". In: *Physical Review Letters* 100.2, 020603 (2008). DOI: 10.1103/PhysRevLett.100.020603.

[10] PLUMED developers. *sum_hills*. URL: https://www.plumed.org/doc-v2.9/user-doc/html/sum_hills.html (visited on 09/21/2026).

[11] PLUMED developers. *METAD: metad.bias, metad.rbias and c(t)*. URL: https://www.plumed.org/doc-v2.9/user-doc/html/_m_e_t_a_d.html (visited on 09/21/2026).

[12] M. Bonomi, A. Barducci, and M. Parrinello. "Reconstructing the equilibrium Boltzmann distribution from well-tempered metadynamics". In: *Journal of Computational Chemistry* 30.11 (2009), pp. 1615–1621. DOI: 10.1002/jcc.21305.

[13] P. Tiwary and M. Parrinello. "A Time-Independent Free Energy Estimator for Metadynamics". In: *The Journal of Physical Chemistry B* 119.3 (2015), pp. 736–742. DOI: 10.1021/jp504920s.

[14] PLUMED developers. *REWEIGHT_METAD*. URL: https://www.plumed.org/doc-v2.9/user-doc/html/_r_e_w_e_i_g_h_t__m_e_t_a_d.html (visited on 09/21/2026).

[15] W. Kabsch and C. Sander. "Dictionary of protein secondary structure: Pattern recognition of hydrogen-bonded and geometrical features". In: *Biopolymers* 22.12 (1983), pp. 2577–2637. DOI: 10.1002/bip.360221211.

[16] A. Rodriguez and A. Laio. "Clustering by fast search and find of density peaks". In: *Science* 344.6191 (2014), pp. 1492–1496. DOI: 10.1126/science.1242072.

[17] P. J. Rousseeuw. "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis". In: *Journal of Computational and Applied Mathematics* 20 (1987), pp. 53–65. DOI: 10.1016/0377-0427(87)90125-7.

[18] C. R. Harris, K. J. Millman, S. J. van der Walt, et al. "Array programming with NumPy". In: *Nature* 585.7825 (2020), pp. 357–362. DOI: 10.1038/s41586-020-2649-2.

[19] P. Virtanen, R. Gommers, T. E. Oliphant, et al. "SciPy 1.0: fundamental algorithms for scientific computing in Python". In: *Nature Methods* 17.3 (2020), pp. 261–272. DOI: 10.1038/s41592-019-0686-2.

[20] F. Pedregosa, G. Varoquaux, A. Gramfort, et al. "Scikit-learn: Machine Learning in Python". In: *Journal of Machine Learning Research* 12.85 (2011), pp. 2825–2830. URL: https://www.jmlr.org/papers/v12/pedregosa11a.html.

[21] Snakemake developers. *Snakefiles and Rules*. URL: https://snakemake.readthedocs.io/en/stable/snakefiles/rules.html (visited on 09/21/2026).

[22] NumPy developers. *Parallel random number generation*. URL: https://numpy.org/doc/stable/reference/random/parallel.html (visited on 09/21/2026).

[23] SciPy developers. *wasserstein_distance*. URL: https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.wasserstein_distance.html (visited on 09/21/2026).

[24] SciPy developers. *RegularGridInterpolator*. URL: https://docs.scipy.org/doc/scipy/reference/generated/scipy.interpolate.RegularGridInterpolator.html (visited on 09/21/2026).

[25] C. J. Geyer. "Practical Markov Chain Monte Carlo". In: *Statistical Science* 7.4 (1992), pp. 473–483. DOI: 10.1214/ss/1177011137.

[26] L. Hubert and P. Arabie. "Comparing partitions". In: *Journal of Classification* 2.1 (1985), pp. 193–218. DOI: 10.1007/BF01908075.

[27] L. Wratten, A. Wilm, and J. Göke. "Reproducible, scalable, and shareable analysis pipelines with bioinformatics workflow managers". In: *Nature Methods* 18.10 (2021), pp. 1161–1168. DOI: 10.1038/s41592-021-01254-9.

[28] M. d'Errico, E. Facco, A. Laio, and A. Rodriguez. "Automatic topography of high-dimensional data sets by non-parametric density peak clustering". In: *Information Sciences* 560 (2021), pp. 476–492. DOI: 10.1016/j.ins.2021.01.010.

[29] S. Sasmal, T. Pal, G. M. Hocky, and M. McCullagh. "Quantifying Unbiased Conformational Ensembles from Biased Simulations Using ShapeGMM". In: *Journal of Chemical Theory and Computation* 20.9 (2024), pp. 3492–3502. DOI: 10.1021/acs.jctc.4c00223.

[30] N. Michaud-Agrawal, E. J. Denning, T. B. Woolf, and O. Beckstein. "MDAnalysis: A toolkit for the analysis of molecular dynamics simulations". In: *Journal of Computational Chemistry* 32.10 (2011), pp. 2319–2327. DOI: 10.1002/jcc.21787.

[31] CINECA. *Leonardo: HPC documentation*. URL: https://docs.hpc.cineca.it/hpc/leonardo.html (visited on 09/21/2026).

A Full parameter table

Iterative density-peak discovery

Within each random group of about 30 000 frames, discovery proceeds as follows.

1. **Generate candidates.** Start discovery at $d_c = 1.3$, rank candidate centres by $\rho_i\delta_i$, and try 10 down to 2 centres.
2. **Check each cluster.** Require at least 300 members and a mean silhouette of at least 0.3 for acceptance.
3. **Choose the candidate.** Keep the one with the most acceptable clusters. A tie retains the earlier, higher-centre-count candidate.
4. **Continue on the remainder.** Remove accepted members and multiply d_c by 0.9 before the next iteration.
5. **Stop discovery.** Stop below 300 remaining points or after five consecutive unsuccessful iterations. Residual frames receive their final labels only in STEP 3.

Table 10 Reproduction settings and execution profiles. These parameters are implemented but do not imply a completed empirical run.

Parameter	Value	Evidence / role
Replicas	3	Primary paper; observed table segmentation
Analysis interval	1–3 µs	Published baseline; inclusive source-labelled ps boundaries
Temperature	310 K	Published weighting parameter; no new MD run
Bias factor γ	10	Published weighting parameter
Molar gas constant	0.008314462618 kJ mol^{-1} K^{-1}	Implementation unit convention
CV conversions	$\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$	Published descriptor scaling
Baseline features	n_α, n_β, R_g, SASA	Pooled min–max normalization
Initial d_c	1.3	Published baseline
Cutoff multiplier	0.9	Published 10% reduction
Center-count scan	10 down to 2	Published baseline
Minimum cluster size	300 frames	Published acceptance criterion
Minimum mean silhouette	0.3	Published acceptance criterion
Failure stopping count	5	Consecutive iterations with no accepted cluster
Distance storage	float32	Source-informed; blockwise construction
Consensus	Average linkage	Marginal Wasserstein norm
Elbow smoothing	$s = 5$	Supplied source heuristic
Elbow search range	First 90% of merges	Supplied source heuristic
Final assignment	One nearest-centroid pass	No iterative refitting
Master seed	42	Default project configuration; recorded per run
Smoke / medium groups	3,000 / 10,000	Reduced execution profiles
Smoke / medium discovery groups	2 / 3	Not full-production evidence

Table 10 continued

Parameter	Value	Evidence / role
Full group size	About 30,000	All groups; full execution disabled
Seed counts	2 smoke; 5 pilot/medium; 20 full default	Full robustness configurable; not performed
Bootstrap draws	500	Conditional moving-block extension; not performed on real weights
Resource policy	Bounded sequential local workers	Full execution held for review

Completed descriptor-only GMM pilot

The protocol was committed before execution. Its four operations separate the choice of fitting observations from the comparisons made afterwards:

1. **Fix the sample.** Select 1,000 frames per replica with seed 20261007. Use the same 3,000 observations for every fit.
2. **Fit the transformation.** Estimate means and population standard deviations on those rows only. Share the transformation of retained descriptors between the four- and three-feature representations.
3. **Fit the models.** For each $K = 2, \ldots, 6$, use seeds 42–46: 50 fits overall. Each fit uses one initialization, at most 500 iterations, tolerance 10^{-3} and covariance regularization 10^{-6}.
4. **Compare assignments.** Label all 300,003 frames, including fitting rows. Use five paired-seed ARIs to compare representations and ten seed pairs to compare initializations within each representation.

The following settings belong to the successful independent experiment, not to the published DPC baseline. The fit sample and initializations were recorded before execution.

Table 11 Settings of the completed GMM experiment, as recorded in its run manifest.

Parameter	Value	Evidence / role
Fit sample	1,000 per replica	3,000 rows; shared across models
Sampling seed	20261007	Local NumPy generator
Feature spaces	Four descriptors; omit SASA	Same scaling for shared features
Scaling	Fit-sample mean and SD	No evaluation-data refit
Components	2, 3, 4, 5, 6	All counts retained in comparison
Initialization seeds	42, 43, 44, 45, 46	Paired across representations
Initializations per fit	1	50 total fitted models
Covariance	Full	Regularization 10^{-6}
Convergence settings	500 iterations; tolerance 10^{-3}	All fits converged numerically
Evaluation	300,003 rows	Includes fit rows; descriptive comparison
Resource limits	600 s; 4 GiB	Sequential local worker

B Mathematical derivations and estimator checks

Gaussian hills and bias construction

The accumulated potential can be written as a sum of local Gaussian hills:

$$V(s,t) = \sum_{t_k<t} h_k \exp\left[-\frac{1}{2}\sum_a \frac{(s_a - s_a(t_k))^2}{\sigma_a^2}\right]. \tag{7}$$

In Equation (7), s is the CV vector, t_k a deposition time, h_k the hill height and σ_a its width along coordinate a. Well-tempered heights follow $h_k = h_0 \exp[-V(s(t_k), t_k)/(k_B \Delta T)]$ in molecular energy units; molar units require R instead of k_B.

Why the sign of the final-bias correction is negative in FES form

Under a static bias V, the observed configuration density is proportional to $\exp[-\beta_T(U+V)]$. Multiplying by $\exp(\beta_T V)$ recovers relative canonical density. The WT relation $V \simeq -(1-1/\gamma)F + C$ therefore gives the negative free-energy sign in Equation (2). Applying $\exp(-\beta_T V)$ would reverse the correction. This derivation assumes that using a final bias is appropriate for the analyzed interval; it is not a proof that the supplied trajectories satisfy that assumption.

Kish effective sample size

Unequal statistical weights require a separate diagnostic:

$$N_{\mathrm{eff,Kish}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{8}$$

Equation (8) measures weight concentration without correcting temporal dependence. It is not automatically combined with correlation ESS. No empirical Kish ESS is available without validated weights.

Equal-replica mixture and cluster moments

Let $\widetilde{w}_{ri}$ sum to one within replica r. The pooled expectation of A is $(1/3)\sum_r \sum_i \widetilde{w}_{ri} A_{ri}$. Cluster relevance uses A equal to a membership indicator. For a conditional cluster mean, divide the pooled weighted sum of the descriptor by the pooled membership mass. Computing three conditional means and averaging them equally instead would answer a different question when cluster populations differ across replicas.

Bilinear interpolation test

For a cell with fractional coordinates $u, v \in [0,1]$, interpolation combines its four corner values with weights $(1-u)(1-v)$, $u(1-v)$, $(1-u)v$, and uv. Their sum is one. Any surface $a + b\alpha + c\beta + d\alpha\beta$ is reproduced exactly within floating-point tolerance. The test suite includes such a mixed-term surface, reordered input rows, boundaries and invalid-grid cases, providing stronger evidence than checking a single grid node.

C Additional diagnostics and interpretation

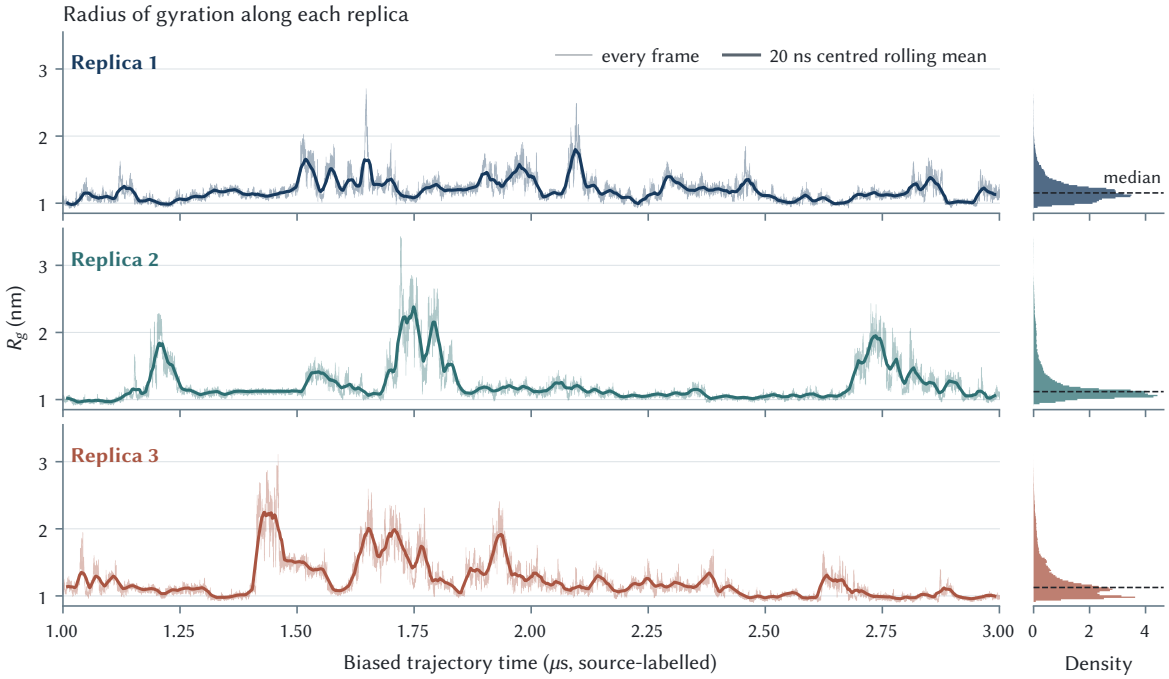


Figure 9 Radius of gyration along the biased trajectory of each replica. Thin lines show every frame and bold lines a centred 20 ns (1,000-frame) rolling mean. The right-hand histograms are unweighted densities with dashed lines at the medians (1.153, 1.118 and 1.126 nm).

Table 12 Agreement between feature sets across the five paired initializations.

Components	Median ablation ARI	Observed range
2	0.690	0.188–0.868
3	0.864	0.862–0.868
4	0.607	0.548–0.622
5	0.531	0.382–0.696
6	0.457	0.294–0.461

Component counts were specified for comparison, not selected as physical numbers of states.

The saved tables contain per-replica raw descriptor counts, means, standard deviations, extrema, common-bin histograms, the full four-feature correlation matrix, PCA transformation parameters and explained variance, and the reported correlation-estimator diagnostics. The ACF figure is a plotting subsample, not the series used to estimate g. No weights are present in these real-data diagnostic summaries.

The half-window mean-shift diagnostic is descriptive. A large shift can arise from slow fluctuations, nonstationarity, or changing occupation of sampled regions. Its threshold is an explicit project heuristic. A low shift alone would not establish convergence, and no threshold outcome can recover an unbiased time scale from WT-MetaD time. Future evaluation should preserve these distinctions when adding weighted estimates.

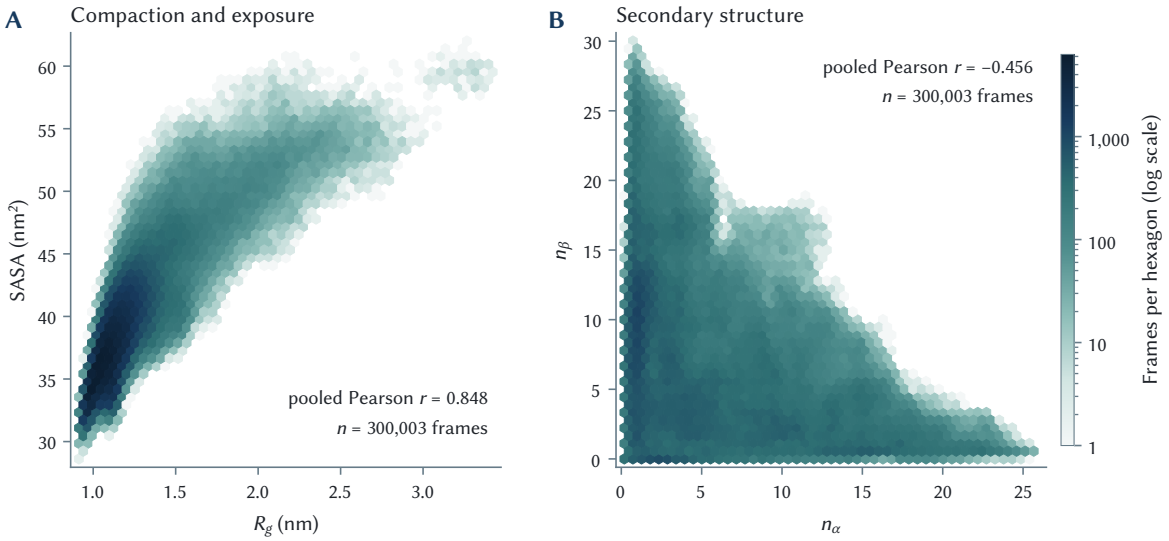


Figure 10 Unweighted descriptor landscape of the pooled sample (300,003 frames, three replicas, 1–3 µs). (A) Radius of gyration against SASA. (B) n_α against n_β. Shading is the log number of frames per hexagonal cell on a shared scale.

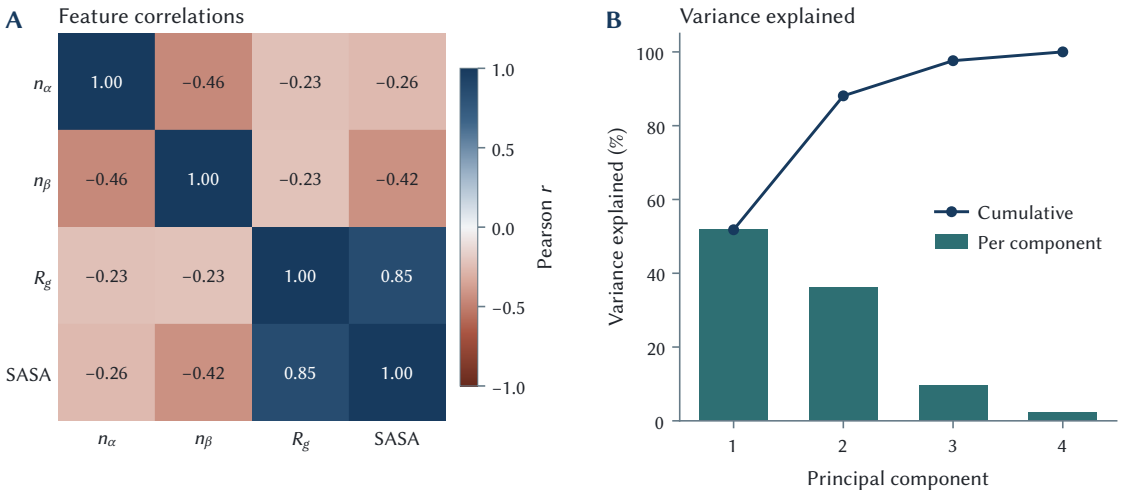


Figure 11 Relationships between the four descriptors in the pooled, unweighted sample. Left: Pearson correlation matrix. Right: variance explained by standardized PCA, with cumulative variance shown by the line.

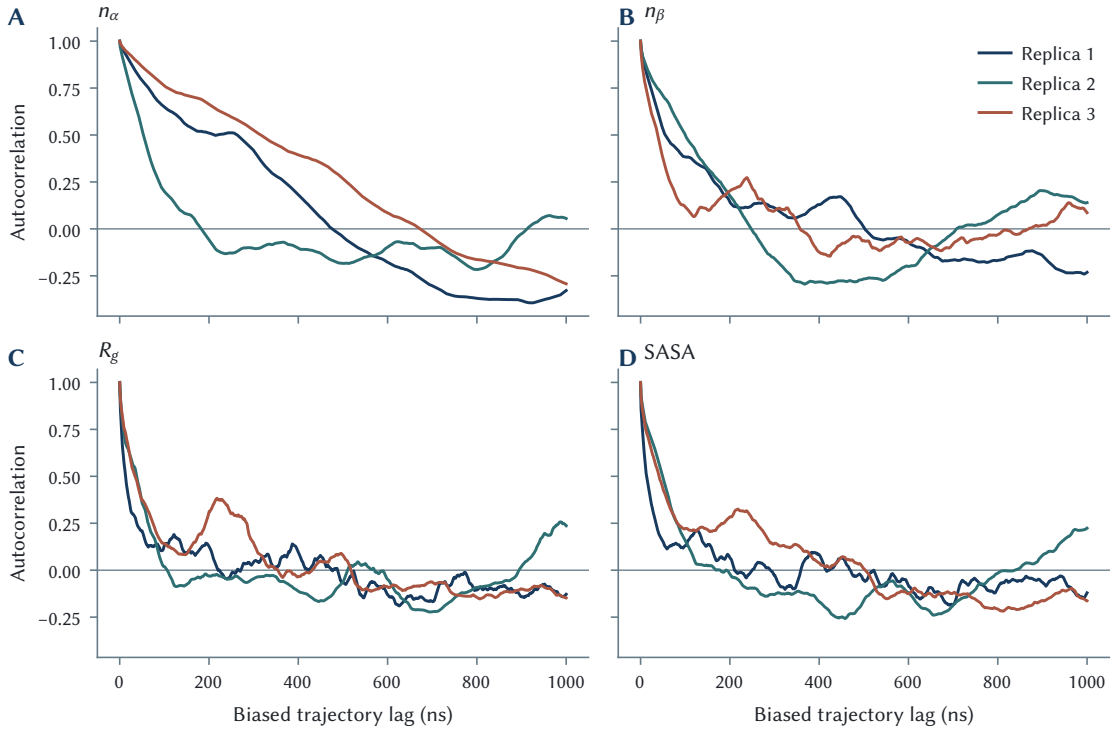


Figure 12 Autocorrelation functions calculated separately for each replica and descriptor. For readability the figure displays a subset of lag points over the first half of the available window; estimation uses the full ACF sequence. The horizontal axis is biased trajectory time.

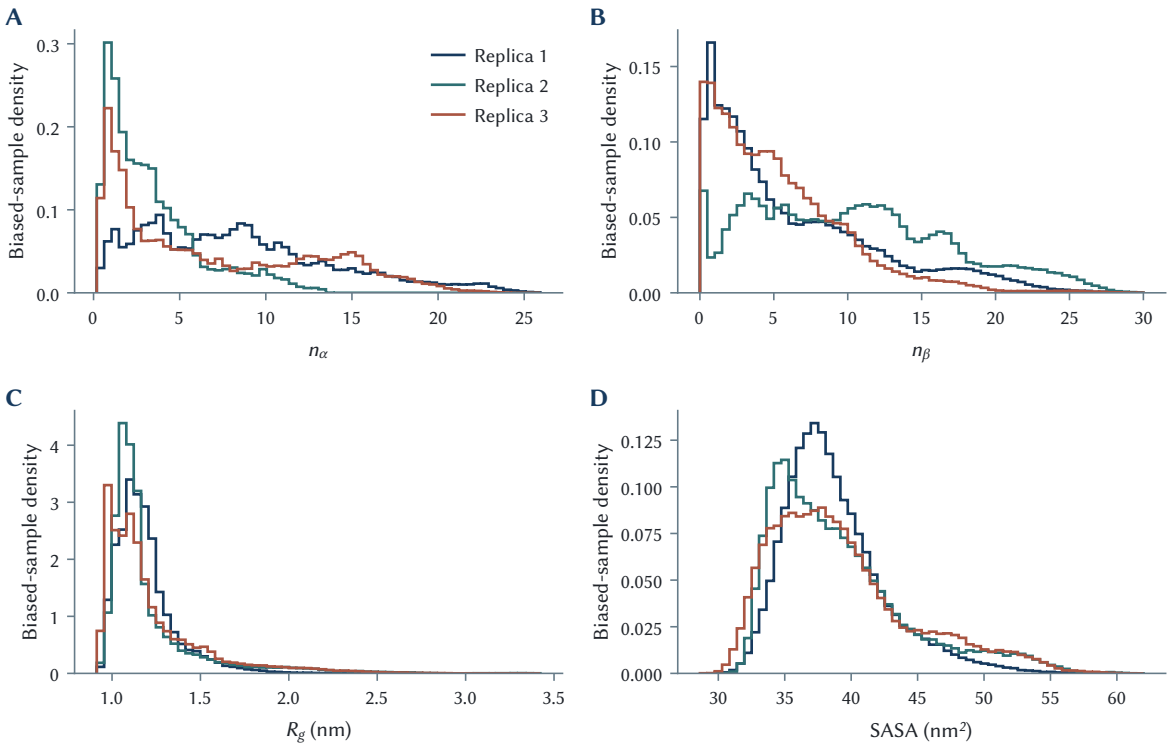


Figure 13 Per-replica distributions of the recorded descriptors. Each descriptor uses common bin edges, with separate density normalization for each replica. The curves retain the simulation bias and are not equilibrium distributions. Units follow the source labels.

Table 13 All recorded correlation estimates, each flagged as unreliable by at least one stated check. g denotes statistical inefficiency. The half-window shift is the absolute difference of means divided by the full-series standard deviation. The time axis belongs to the biased trajectory.

Replica	Descriptor	g	τ_{int} (ns)	$N_{\mathrm{eff,corr}}$	Half shift / SD
1	n_α	22067	220.7	4.5	1.28
1	n_β	11079	110.8	9.0	0.83
1	R_g	3679	36.8	27.2	0.27
1	SASA	3490	34.9	28.7	0.36
2	n_α	6259	62.6	16.0	0.37
2	n_β	10843	108.4	9.2	0.41
2	R_g	3951	39.5	25.3	0.33
2	SASA	5325	53.2	18.8	0.17
3	n_α	31742	317.4	3.2	1.61
3	n_β	5880	58.8	17.0	0.71
3	R_g	6259	62.6	16.0	0.88
3	SASA	9958	99.6	10.0	0.87

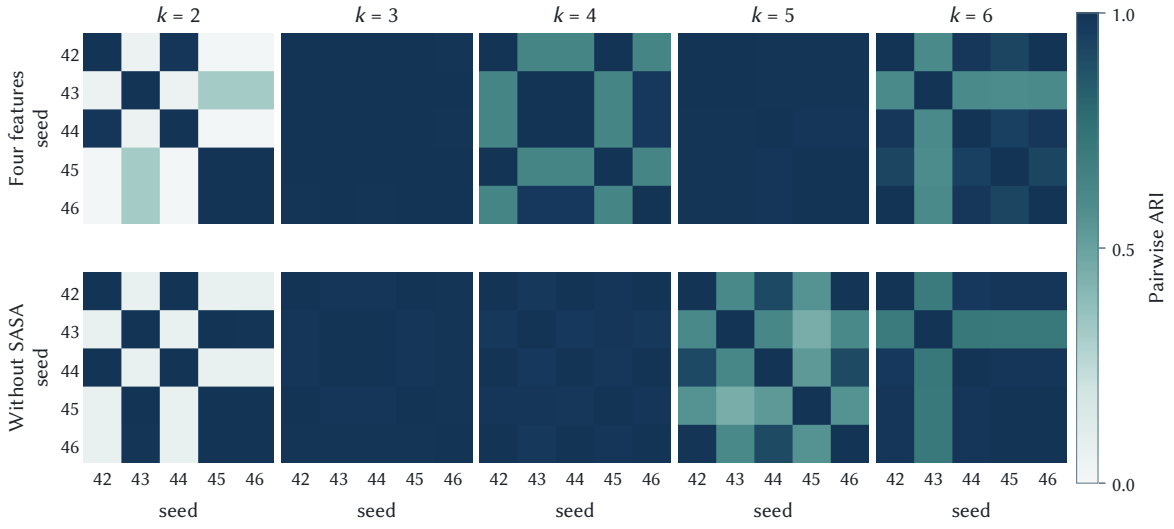


Figure 14 All pairwise seed ARIs from the GMM pilot. Columns show each tested component count; rows distinguish the four-feature and SASA-ablated representations. The diagonal is identity. All fits converged, so no entries are omitted. Similar numerical stopping status can coexist with different partitions.

D Code audit summary

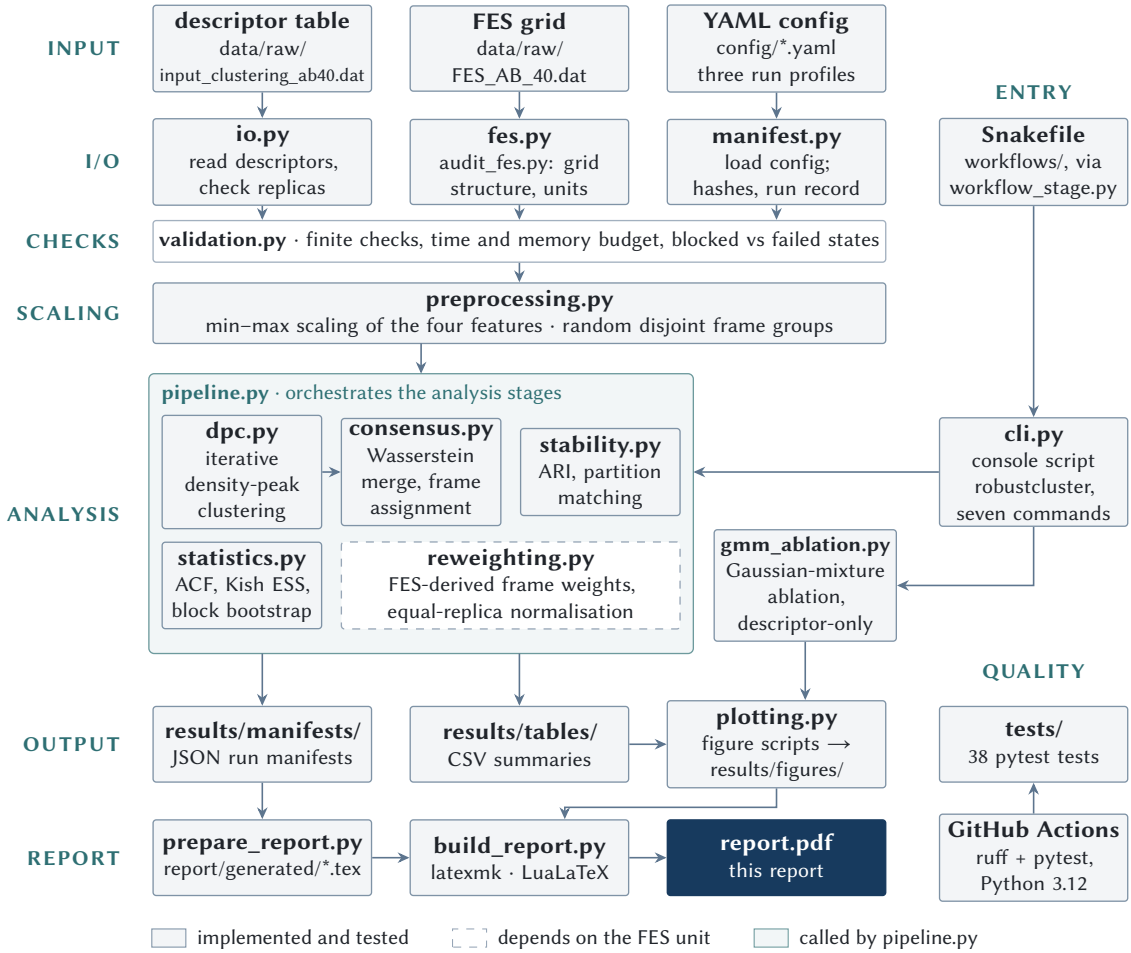


Figure 15 Architecture of the `robustcluster-abeta` package. Inputs pass through validation and scaling to the analysis modules orchestrated by `pipeline.py`; results go to manifest-backed tables and figures and then into this report. `gmm_ablation.py` is the separate descriptor-only experiment of Section 5.5. The dashed module depends on the FES unit and was run under the assumption of Section 6.5.

Two numerical tests

Two small examples from the 38-test suite show how a check can identify a numerical error without relying on the output of the full pipeline.

Interpolation with a known answer. An analytic bilinear surface includes a mixed-coordinate term, so a query between nodes tests both axes together. Reordering its rows must preserve the interpolated answer. Deleting or duplicating a grid point, or querying outside the bounds, must instead raise an error. The same fixture therefore checks both a valid calculation and the rejection of inputs for which that calculation would be inappropriate.

Normalization with an invariant. An arbitrary additive FES constant must disappear after weight normalization. A second check gives two replicas unequal numbers of frames and requires

each to contribute half of the pooled mass. This catches pooling by row count, which would look correct on three equal-length replicas. Other weight tests check the formula's sign and weighted moments.

The remaining checks cover parsing, scaling, residual exclusion and final assignment. Renaming cluster labels must leave partition agreement unchanged; changing evaluation extremes must not alter a scaler fitted on fixed training rows. These checks isolate failures that could otherwise be mistaken for a change in the clustering method.

A synthetic end-to-end fixture exercises discovery, consensus, weighting, summaries and figure generation. Synthetic GMM cases check repeatability and missing-family handling. An autoregressive fixture probes the correlation estimator; an archive test prevents an unrelated enclosing Git repository from supplying false provenance. The fixtures establish software behaviour on known inputs (Scope 5).

The project inventory records immutable source hashes, table dimensions and missing inputs. The method map links each step to paper evidence, source behavior, the modular implementation and validation tests. The decision log records scientific category and expected effect. The open-questions register prevents uncertain provenance from being converted into hidden defaults.

Scientific V2 is absent, so the audit contains no invented intermediate diff. The supplied original remains excluded from execution and lint collection because its syntax error and top-level writes are part of the preserved evidence. New code is tested independently. Public source packaging excludes the raw table and original script until their redistribution rights are established.

The checked bibliography contains journal articles and official documentation records; its matrix links each citation to its question, method, relevance, and report section. Online-first dates and issue years are distinguished. References whose publisher verification remained unresolved are documented outside the BibTeX file rather than filled from memory.

E Reproducibility instructions and software environment

The project source, tests, configurations and this report are available at RobustCluster-Aβ on GitHub. The Python package and LaTeX sources can be inspected separately or downloaded together using GitHub's *Code / Download ZIP* option. The repository guide distinguishes code verification and report regeneration from analyses requiring restricted inputs.

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-lock.txt
python -m pip install --no-deps -e .
pytest -q
ruff check src tests scripts
robustcluster validate --config config/paper_reproduction.yaml
robustcluster diagnostics --config config/robustness.yaml
robustcluster cluster --config config/paper_reproduction_full.yaml --profile full
robustcluster reproduce --config config/paper_reproduction_full_assumed.yaml --profile full
```

The raw file must first be obtained and checked against its documented SHA256. The standalone commands and Snakemake workflow use the same package functions. The report imports successful diagnostics, FES-audit, GMM and figure manifests with matching hashes. Its generated editorial artwork and real browser screenshots have separate provenance records. LuaLaTeX, Biber and latexmk build the supplied report sources; current paths and tool versions are recorded in the build manifest.

The full-profile configurations record the measured resource estimates that justified enabling them (about 15 minutes and 4 GiB per clustering run). The configuration with the assumed FES unit states that assumption in its header and in every manifest it produces; once the unit is confirmed, only its two FES fields need to change. Outputs that would redistribute the FES or per-frame descriptor values stay local, with their hashes kept in the manifests. No calculation in this report used an HPC system [31].

Software environment

Table 14 Software versions for the reported local descriptor diagnostics.

Component	Version in diagnostic manifest
Python	3.12.6
numpy	2.5.3
scipy	1.18.1
scikit-learn	1.9.1
pandas	2.3.3
matplotlib	3.11.2
robustcluster-abeta	0.1.0

The GMM pilot manifest records the same versions. They do not describe the original authors' simulation environment. The complete dependency lock and report-tool versions are separate reproducibility artifacts.

F Personal projects and source-code evidence

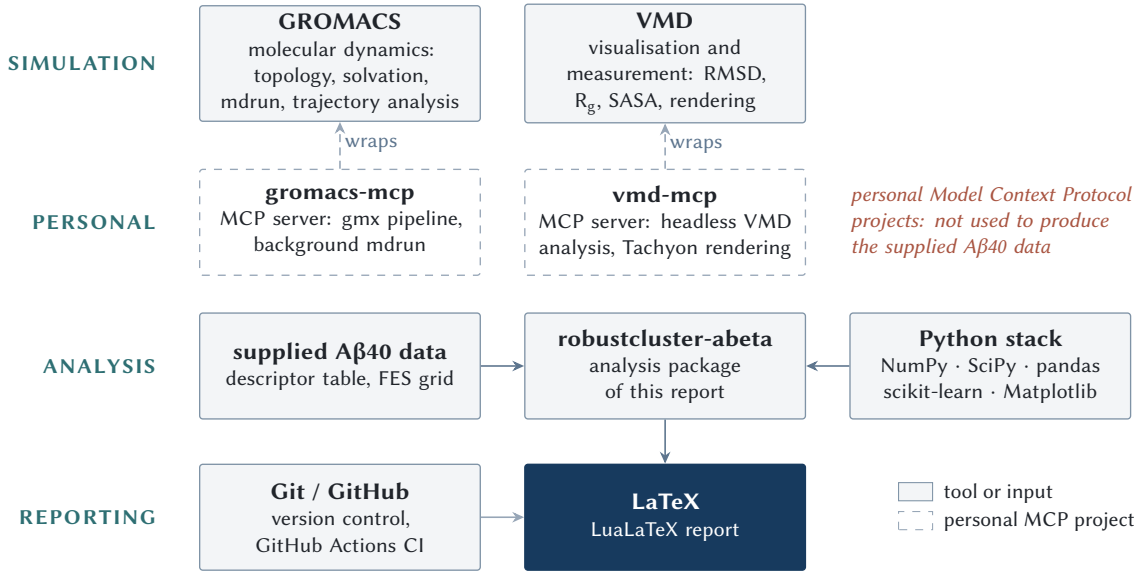


Figure 16 Tools used during the internship. GROMACS and VMD belong to the simulation block; the Python stack, Git/GitHub and LaTeX produced the analysis and this report. Dashed boxes are personal Model Context Protocol servers, which were not used to produce the supplied Aβ40 data.

VMD MCP: frame measurements

The inspected source constructs Tcl commands for measurement and rendering. Its gyration function passes an atom selection and mass-weighting option to VMD, then returns frame-indexed values.

GROMACS MCP: execution interface

The inspected source includes preparation and analysis helpers. Its background-run interface checks paths and records output in a log.

The following images are genuine browser captures from public, commit-specific GitHub source pages on 7 October 2026. They document implementation, not simulation execution. The full images are preserved unchanged; short accompanying searchable listings are a separate presentation format.

02 Measure each saved frame VMD MCP / source lines 250–266

```
    weight = "weight mass" if mass_weighted else ""
    tcl = f"""
{_load_block(structure, trajectory)}
set sel [atomselect $mol "{_tcl_quote(selection)}"]
set n [molinfo $mol get numframes]
for {{set i 0}} {{$i < $n}} {{incr i}} {{
  $sel frame $i
  puts "{MARKER} frame=$i rgyr=[measure rgyr $sel {weight}]"
}}
quit
"""
    res = _run_tcl(tcl)
    series = [
        {"frame": int(m["frame"]), "rgyr": float(m["rgyr"])}
        for m in res.get("markers", [])
        if "rgyr" in m
    ]
```

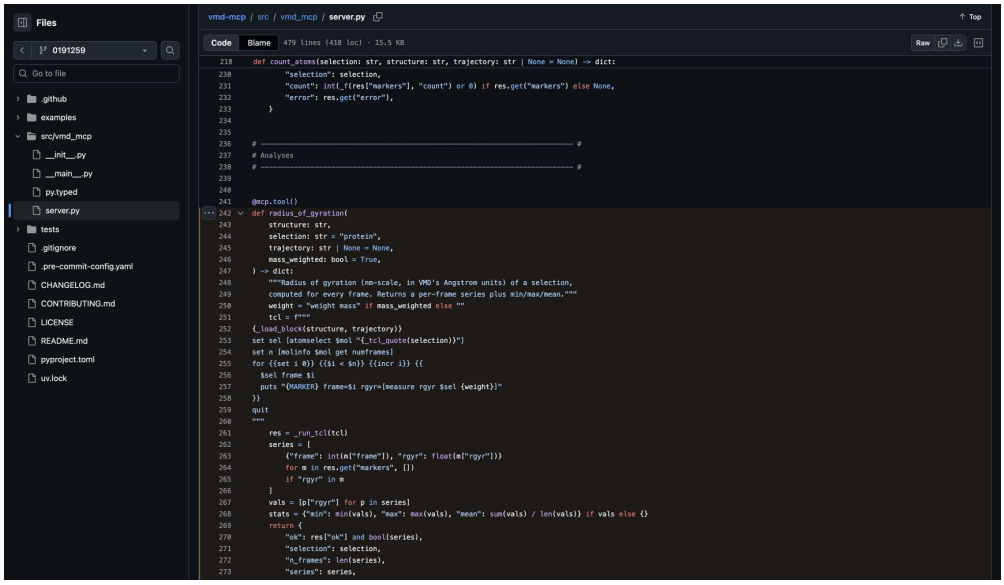


Figure 17 VMD MCP source at commit 0191259: the gyration wrapper constructs a Tcl frame loop and parses marked values. The screenshot shows the repository tree, source path and real line numbers. The function's documented unit convention must be checked before comparison with other tools. Open the inspected source.

03 Validate before starting a job GROMACS MCP / source lines 1452–1467

```
1452        wd, err = _workdir_or_error(workdir)
1453        if err:
1454            return err
1455        assert wd is not None
1456        if ntomp is not None and ntomp <= 0:
1457            return _error("ntomp must be positive when provided")
1458        if nsteps is not None and nsteps < 0:
1459            return _error("nsteps must be non-negative when provided")
1460        safe_tpr = _safe_file_arg(wd, tpr, "tpr", must_exist=True)
1461        safe_deffnm = _safe_deffnm(deffnm)
1462        args = [GMX_BIN, "mdrun", "-v", "-s", safe_tpr, "-deffnm", safe_deffnm]
1463        if ntomp is not None:
1464            args += ["-ntomp", str(ntomp)]
1465        if nsteps is not None:
1466            args += ["-nsteps", str(nsteps)]
1467        args += _validate_arg_tokens(extra_args, "extra_args")
```

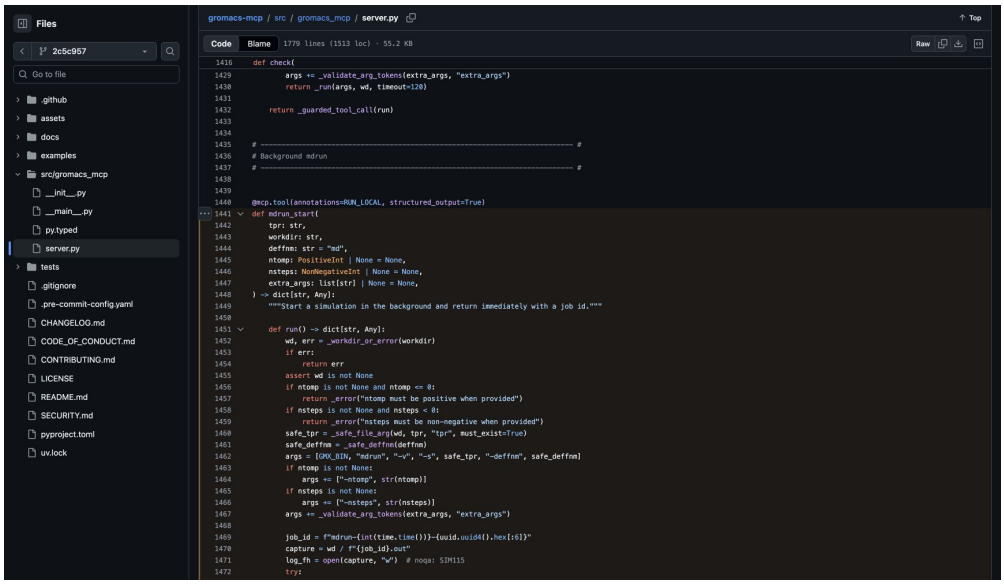


Figure 18 GROMACS MCP source at commit `2c5c957`: the background-run wrapper validates inputs, constructs the `mdrun` command and prepares an output log. This source view does not establish that an Aβ40 simulation was launched through the tool. Open the inspected source.